

Characteristic Classes of Vector and Principal Bundles

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Prerequisites. Smooth manifolds, differential forms and de Rham cohomology, and the elementary theory of Lie groups and their Lie algebras (recalled in the preliminary section below). The exposition is otherwise self-contained.

Introduction

A *characteristic class* of a vector bundle is a cohomology class of the base space, functorially attached to the bundle, which measures the extent to which the bundle fails to be trivial. The subject has a curious dual nature: characteristic classes are *topological* invariants — they depend only on the isomorphism class of the bundle, not on any auxiliary choice — yet they are computed from *geometric* data, namely from the curvature of a connection. The bridge between the two points of view is the Chern–Weil theory, which we develop in Chapter 3.

The theory was created between 1935 and 1950 in response to concrete geometric problems. Stiefel (1935) and Whitney (1937) introduced the classes that now bear Whitney’s name in order to obstruct the existence of linearly independent tangent vector fields on a manifold. Pontryagin (1942) and Chern (1946) found the integral classes associated with real and complex bundles, and Chern’s construction, via the curvature of a connection, is the template for everything that follows. The Euler class is even older in spirit: it is the globalisation of the Poincaré–Hopf index theorem, and its integrated form is the Gauss–Bonnet theorem. The modern cohomological formulation — classifying spaces and universal characteristic classes — is due to Milnor and was systematised in Milnor–Stasheff’s classic book.

These notes are organised as follows. Chapter 1 recalls vector bundles and introduces connections and curvature. Chapter 2 develops the theory of principal bundles and their connections, which is the natural framework for Chern–Weil theory in Chapter 3. Chapters 4–7 treat the four families of characteristic classes: Chern classes, Pontryagin classes, the Euler class, and Stiefel–Whitney classes. Chapter 8 presents the classifying-space point of

view and the universal characteristic classes. A short final chapter collects the connections with the index theorems and traces the historical arc of the subject.

Throughout, all manifolds, bundles, maps, sections and connections are understood to be smooth (C^∞), and all vector spaces are over $\mathbb{R}$ or $\mathbb{C}$ as indicated. The de Rham cohomology of a manifold M is denoted $H_{\text{dR}}^*(M)$.

Notation and Conventions

The following conventions are fixed once and for all.

- M is a smooth manifold of dimension m ; $\Omega^k(M) = \Omega^k(M; \mathbb{R})$ is the space of smooth k -forms; $\Omega^k(M; V)$ the space of V -valued k -forms.
- $E \rightarrow M$ is a smooth (real or complex) vector bundle of rank r ; $\Gamma(E)$ denotes its space of smooth sections; $\text{End } E = \text{Hom}(E, E)$.
- A *frame* of E over $U \subseteq M$ is an ordered collection $e = (e_1, \dots, e_r)$ of sections forming a basis of each fibre; the frame determines a trivialisation $E|_U \cong U \times \mathbb{F}^r$.
- $\text{GL}(r; \mathbb{F})$ is the general linear group; for $\mathbb{F} = \mathbb{R}$ we often write $\text{GL}(r)$.
- The wedge product of matrix-valued forms is computed with ordinary matrix multiplication of the coefficients: if $\omega = (\omega_j^i)$ and $\eta = (\eta_j^i)$, then $(\omega \wedge \eta)_j^i = \sum_k \omega_k^i \wedge \eta_j^k$.
- Summation over repeated upper and lower indices is understood.
- A *connection* is written ∇ ; its curvature is Ω or R .
- Cohomology is de Rham cohomology with real or complex coefficients unless otherwise stated; $\mathbb{Z}$ and $\mathbb{Z}/2$ coefficients are used in Chapters 7 and 8.
- For a complex vector bundle E , the notation $\bar{E}$ denotes the conjugate bundle and $E_{\mathbb{R}}$ the underlying real bundle.

Preliminaries on Lie Groups and Lie Algebras

This section recalls, with a minimum of machinery, the Lie-theoretic notions used freely from Chapter 2 onwards: Lie groups and their Lie algebras, the exponential map, the adjoint representations, and the Maurer–Cartan form. Its purpose is only to fix notation and make the few facts we need self-contained; nothing here is new. Throughout, “Lie group” means a real Lie group, except where the complex groups $\text{GL}(n; \mathbb{C})$ and $\text{U}(n)$ are named explicitly.

Lie groups. A *Lie group* is a smooth manifold G together with smooth group operations — a multiplication $G \times G \rightarrow G$, $(g, h) \mapsto gh$, and an inversion $G \rightarrow G$, $g \mapsto g^{-1}$ — making G into a group. Equivalently, G is a group object in the category of smooth manifolds. Every group needed in these notes is a *matrix group*, that is, a closed subgroup of $\text{GL}(n; \mathbb{R})$ or $\text{GL}(n; \mathbb{C})$:

$$\text{GL}(n; \mathbb{R}), \quad \text{GL}(n; \mathbb{C}), \quad \text{O}(n), \quad \text{SO}(n), \quad \text{U}(n), \quad \text{SU}(n), \quad \text{Sp}(n),$$

the general linear, orthogonal, special orthogonal, unitary, special unitary and symplectic groups. A closed subgroup of a Lie group is itself a Lie group. For $g \in G$, left and right translation are the diffeomorphisms $L_g(h) = gh$ and $R_g(h) = hg$.

Lie algebras. The *Lie algebra* of G is the tangent space to G at the identity e :

$$\mathfrak{g} = T_e G.$$

Each $X \in \mathfrak{g}$ extends uniquely to a *left-invariant vector field* $\tilde{X}$ on G , defined by $\tilde{X}_g = (L_g)_* X$; conversely, a left-invariant vector field is determined by its value at e . Since the Lie bracket of left-invariant vector fields is again left-invariant, it descends to a bilinear bracket

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}, \quad (X, Y) \mapsto [X, Y],$$

which is antisymmetric and satisfies the Jacobi identity $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$; thus $\mathfrak{g}$ is a Lie algebra in the abstract sense.

For a matrix group $G \subseteq \mathrm{GL}(n; \mathbb{F})$ the Lie algebra is a subspace of the matrix algebra $M_n(\mathbb{F})$, and the bracket is the commutator $[X, Y] = XY - YX$. The Lie algebras occurring in these notes are

$$\begin{aligned} \mathfrak{gl}(n; \mathbb{F}) &= M_n(\mathbb{F}), & \mathfrak{so}(n) &= \{ X \in M_n(\mathbb{R}) : X + X^T = 0 \}, \\ \mathfrak{u}(n) &= \{ X \in M_n(\mathbb{C}) : X + X^* = 0 \}, & \mathfrak{su}(n) &= \{ X \in \mathfrak{u}(n) : \mathrm{tr} X = 0 \}; \end{aligned}$$

thus $\mathfrak{so}(n)$ consists of the skew-symmetric and $\mathfrak{u}(n)$ of the skew-Hermitian matrices. Informally, $\mathfrak{g}$ is the “infinitesimal” or “linearised” form of G : its elements are the velocities of curves through e , that is, the directions in which one can start moving inside G .

The exponential map. Every $X \in \mathfrak{g}$ integrates to a unique *one-parameter subgroup* $\gamma_X : \mathbb{R} \rightarrow G$, a homomorphism with $\gamma_X(0) = e$ and $\gamma'_X(0) = X$. The *exponential map* is

$$\exp : \mathfrak{g} \rightarrow G, \quad \exp(X) = \gamma_X(1).$$

It satisfies $\exp((s+t)X) = \exp(sX)\exp(tX)$ and $\frac{d}{dt}\big|_{t=0} \exp(tX) = X$. For a matrix group $\exp$ is the ordinary matrix exponential

$$\exp(X) = e^X = \sum_{k \geq 0} \frac{X^k}{k!},$$

whose convergence in $M_n(\mathbb{F})$ is absolute. The map $\exp$ restricts to a diffeomorphism from a neighbourhood of $0 \in \mathfrak{g}$ onto a neighbourhood of $e \in G$; in particular $t \mapsto \exp(tX)$ is the flow through e of the left-invariant vector field $\tilde{X}$.

The adjoint representations. Conjugation $c_g(h) = ghg^{-1}$ fixes e and therefore differentiates to a linear automorphism of $T_e G = \mathfrak{g}$:

$$\mathrm{Ad}_g = (c_g)_* e \in \mathrm{GL}(\mathfrak{g}), \quad \mathrm{Ad} : G \rightarrow \mathrm{GL}(\mathfrak{g}), \quad g \mapsto \mathrm{Ad}_g.$$

For a matrix group this is $\mathrm{Ad}_g X = gXg^{-1}$. Differentiating Ad in g at the identity gives the adjoint representation of the algebra,

$$\mathrm{ad}_X Y = [X, Y], \quad \mathrm{ad}_X = \frac{d}{dt}\bigg|_{t=0} \mathrm{Ad}_{\exp(tX)}.$$

This last identity is precisely what Chapter 2 uses to differentiate Ad-equivariance conditions along $g = \exp(tA)$. One also has the global form

$$\mathrm{Ad}_{\exp(X)} = e^{\mathrm{ad}_X} = \sum_{k \geq 0} \frac{(\mathrm{ad}_X)^k}{k!},$$

which for matrix groups is the familiar identity $e^X Y e^{-X} = \sum_{k \geq 0} \frac{1}{k!} (\text{ad}_X)^k Y$.

Fundamental vector fields. Let G act smoothly on the right on a manifold P , written $(p, g) \mapsto p \cdot g$. Each $X \in \mathfrak{g}$ determines the *fundamental vector field* $X^\#$ on P by

$$X_p^\# = \left. \frac{d}{dt} \right|_{t=0} p \cdot \exp(tX),$$

the velocity at p of the orbit through p in the direction X . The map $X \mapsto X^\#$ is linear and, when the action is free, injective, with image the directions tangent to the orbits. These are the vertical directions of Chapter 2, and the definition of a connection form is phrased in terms of them.

The Maurer–Cartan form. On a Lie group there is a canonical $\mathfrak{g}$ -valued 1-form, the *Maurer–Cartan form* θ , defined by

$$\theta_g(v) = (L_{g^{-1}})_* v \in \mathfrak{g},$$

so that θ is the left-invariant $\mathfrak{g}$ -valued form with $\theta_e = \text{id}_{\mathfrak{g}}$. For a matrix group it is written $\theta = g^{-1}dg$. It satisfies the *Maurer–Cartan equation*

$$d\theta + \frac{1}{2}[\theta, \theta] = 0, \quad \text{where} \quad [\theta, \theta](u, v) = 2[\theta(u), \theta(v)].$$

This equation is the infinitesimal form of the group law; its curvature-flavoured generalisation, the structure equation $\Omega = d\omega + \frac{1}{2}[\omega, \omega]$ of a connection form, is the starting point of Chern–Weil theory in Chapter 3.

Conventions. The Lie algebra of a Lie group G is always written $\mathfrak{g}$ (Fraktur); the bracket is $[\cdot, \cdot]$; the adjoint representations are Ad (of G) and ad (of $\mathfrak{g}$); the exponential map is $\exp$. The structure groups occurring in these notes are $\text{GL}(n; \mathbb{R})$, $\text{GL}(n; \mathbb{C})$, $\text{O}(n)$, $\text{SO}(n)$, $\text{U}(n)$, $\text{SU}(n)$ and $\text{Sp}(n)$, with Lie algebras as displayed above.

1 Vector Bundles and Connections

1.1 Vector bundles

Definition 1 (Vector bundle). Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. A *vector bundle of rank r* over M is a smooth map $\pi : E \rightarrow M$ together with the structure of an r -dimensional $\mathbb{F}$ -vector space on each fibre $E_p = \pi^{-1}(p)$, satisfying the local triviality condition: every $p \in M$ admits a neighbourhood U and a diffeomorphism $\varphi : \pi^{-1}(U) \rightarrow U \times \mathbb{F}^r$ such that φ maps each fibre E_q linearly onto $\{q\} \times \mathbb{F}^r$. Equivalently, in symbols:

$$\pi : E \rightarrow M, \quad E_p := \pi^{-1}(p) \text{ an } \mathbb{F}\text{-vector space } (\forall p \in M),$$

and for every $p \in M$ there are $U \ni p$ and a diffeomorphism $\varphi : \pi^{-1}(U) \xrightarrow{\cong} U \times \mathbb{F}^r$ whose restriction $\varphi|_{E_q} : E_q \xrightarrow{\cong} \{q\} \times \mathbb{F}^r$ is $\mathbb{F}$ -linear for every $q \in U$. A *section* of E is a smooth map $s : M \rightarrow E$ with $\pi \circ s = \text{id}_M$; the space of sections is

$$\Gamma(E) = \{s \in C^\infty(M, E) : \pi \circ s = \text{id}_M\}.$$

Two local trivialisations $\varphi_\alpha, \varphi_\beta$ over U_α, U_β give rise to a *transition function*

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow \text{GL}(r; \mathbb{F}), \quad \varphi_\alpha \circ \varphi_\beta^{-1}(q, v) = (q, g_{\alpha\beta}(q)v),$$

and the transition functions satisfy the *cocycle condition* $g_{\alpha\beta}g_{\beta\gamma}g_{\alpha\gamma}^{-1} = 1$. Conversely, any such cocycle glues a vector bundle. A *bundle map* (or *homomorphism*) $\phi : E \rightarrow F$ over M is a smooth map with $\pi_F \circ \phi = \pi_E$ which is linear on each fibre; an isomorphism is a bundle map with a two-sided inverse. The pullback f^*E of E along a smooth map $f : N \rightarrow M$ has fibres $(f^*E)_q = E_{f(q)}$.

Example 2 (Tangent bundle and tensor bundles). The tangent bundle $TM \rightarrow M$ is the prototype of a real vector bundle. Its dual T^*M , the tensor bundles $T^{(p,q)}M$ and the exterior powers $\bigwedge^k T^*M$ are obtained functorially; a Riemannian metric is a section of $T^*M \otimes T^*M$ which is positive definite and symmetric in every fibre.

The following elementary operations will be used without comment: direct sum $E \oplus F$, tensor product $E \otimes F$, dual E^* , Hom-bundle $\text{Hom}(E, F) \cong E^* \otimes F$, exterior powers $\bigwedge^k E$, and for complex bundles the conjugate $\bar{E}$. All are defined fibrewise, and the transition functions are the corresponding operations on the transition functions of the factors.

Example 3 (Complexification and realification). For a real bundle E , the *complexification* $E_{\mathbb{C}} = E \otimes_{\mathbb{R}} \mathbb{C}$ has the same transition functions regarded as complex. Its conjugate is canonically isomorphic to itself: $E_{\mathbb{C}} \cong \bar{E}_{\mathbb{C}}$ via $v \otimes z \mapsto v \otimes \bar{z}$; this forces $c_{2k+1}(E_{\mathbb{C}}) = 0$ up to 2-torsion and is the root of the Pontryagin construction in Chapter 5.

Conversely, a complex bundle E has an underlying real bundle $E_{\mathbb{R}}$ (forget the complex structure, real rank doubled). The composite operations satisfy

$$(E_{\mathbb{R}})_{\mathbb{C}} \cong E \oplus \bar{E}, \quad (E_{\mathbb{C}})_{\mathbb{R}} \cong E \oplus E.$$

These two identities are the formal backbone of the relations between real and complex characteristic classes (p_k versus c_{2k}) used throughout.

Remark 4 (Stable equivalence). Two bundles E, F are *stably equivalent* if $E \oplus \varepsilon^k \cong F \oplus \varepsilon^\ell$ for some trivial bundles. Characteristic classes are invariants of the stable equivalence class (a trivial summand contributes nothing), which is why they vanish on stably trivial bundles such as TS^n and why they are naturally packaged in the K -theory of Chapter 4.

1.2 Connections on vector bundles

A connection is a rule for differentiating sections of E along vector fields of M . The abstract definition is the following.

Definition 5 (Connection). A *connection* on $E \rightarrow M$ is an $\mathbb{R}$ -bilinear map

$$\nabla : \mathfrak{X}(M) \times \Gamma(E) \rightarrow \Gamma(E), \quad (X, s) \mapsto \nabla_X s,$$

satisfying, for all $f \in C^\infty(M)$, $X \in \mathfrak{X}(M)$, $s \in \Gamma(E)$,

- (i) C^∞ -linearity in X : $\nabla_{fX}s = f \nabla_X s$;
- (ii) the Leibniz rule in s : $\nabla_X(fs) = X(f)s + f \nabla_X s$.

The section $\nabla_X s$ is the *covariant derivative* of s along X .

Condition (i) says $\nabla_X s$ depends only on the point value X_p ; condition (ii) says ∇ is a first-order differential operator in s . A connection on E extends uniquely to a derivation of the whole tensor algebra of E and E^* , commuting with contractions; in particular it acts on E -valued forms.

Proposition 6 (Connections form an affine space). The space of connections on E is non-empty and is an affine space over $\Omega^1(M; \text{End } E)$: if ∇, ∇' are connections, then $\nabla' - \nabla \in \Omega^1(M; \text{End } E)$, and $\nabla + \alpha$ is a connection for every $\alpha \in \Omega^1(M; \text{End } E)$.

Proof. Both $\nabla_X s$ and $\nabla'_X s$ are C^∞ -linear in X and satisfy the Leibniz rule in s , and these properties are preserved under subtraction; hence the difference $A(X, s) = \nabla'_X s - \nabla_X s$ is C^∞ -linear in X and in s . Fixing X , the map $s \mapsto A(X, s)$ is thus a bundle endomorphism, which we denote $A(X) \in \Gamma(\text{End } E)$; since $A(X, s)$ is C^∞ -linear in X , the assignment $X \mapsto A(X)$ is a smooth section of $\text{Hom}(TM, \text{End } E)$, i.e. an $\text{End } E$ -valued 1-form.

Conversely, for $\alpha \in \Omega^1(M; \text{End } E)$ and a connection ∇ , the sum $\tilde{\nabla} = \nabla + \alpha$ is again a connection: it is C^∞ -linear in X by construction, and for $f \in C^\infty(M)$,

$$\tilde{\nabla}_X(fs) = \nabla_X(fs) + \alpha(X)(fs) = X(f)s + f\nabla_X s + f\alpha(X)s = X(f)s + f\tilde{\nabla}_X s,$$

using the Leibniz rule for ∇ and the C^∞ -linearity of $\alpha(X)$ in the fibre variable. Thus the connections form an affine space over $\Omega^1(M; \text{End } E)$, with $\tilde{\nabla} - \nabla = \alpha$.

It remains to see that the space is non-empty. On a trivialising open set U with local frame $e_1, \dots, e_r$, the *trivial connection*, defined on $s = s^i e_i$ by $\nabla_X^U s = X(s^i) e_i$, is plainly C^∞ -linear in X and satisfies the Leibniz rule. Now choose a locally finite trivialising cover $\{U_\alpha\}$ and a subordinate partition of unity $\{\rho_\alpha\}$, and set

$$\nabla_X s = \sum_\alpha \rho_\alpha \nabla_X^{U_\alpha} s.$$

Each summand is C^∞ -linear in X and obeys the Leibniz rule, hence so does the (locally finite, hence well-defined) sum; moreover $\sum_\alpha \rho_\alpha = 1$ guarantees that ∇ is a globally defined operator on sections of E , independent of the choices of the local frames used to define the ∇^{U_α} . Therefore ∇ is a connection on E . ■

Construction 7 (Local expression). Let $e = (e_1, \dots, e_r)$ be a local frame over U , and write $s = s^i e_i$. Define the *connection 1-form* $\omega = (\omega_j^i)$ by

$$\nabla e_j = \omega_j^i \otimes e_i, \quad \text{i.e.} \quad \nabla_X e_j = \omega_j^i(X) e_i.$$

Then for $s = s^i e_i$,

$$\nabla s = (ds^i + \omega_j^i s^j) \otimes e_i, \quad \text{so that} \quad \nabla_X s = (X(s^i) + \omega_j^i(X) s^j) e_i.$$

Under a change of frame $e' = e g$, i.e. $e'_j = e_i g_j^i$ with $g = (g_j^i) : U \rightarrow \text{GL}(r; \mathbb{F})$, the connection form transforms by

$$\omega' = g^{-1} \omega g + g^{-1} dg. \tag{1}$$

The non-homogeneous term $g^{-1} dg$ shows that ω is *not* a tensor; it is the appearance of this term that underlies gauge theory.

1.3 Curvature

Definition 8 (Curvature). The *curvature* of a connection ∇ on E is the $\mathbb{R}$ -bilinear map

$$R : \mathfrak{X}(M) \times \mathfrak{X}(M) \times \Gamma(E) \rightarrow \Gamma(E),$$

$$R(X, Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X, Y]}s.$$

Proposition 9 (Curvature is a tensor). R is C^∞ -multilinear in X, Y and s , hence defines an element of $\Omega^2(M; \text{End } E)$: the $\text{End } E$ -valued 2-form

$$R = R(X, Y) = (\nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X, Y]}).$$

Proof. It suffices to check C^∞ -linearity in s and in X (then symmetry in X, Y plus skew-symmetry gives the Y -variable). For s :

$$R(X, Y)(fs) = \nabla_X(Y(f)s + f\nabla_Y s) - \nabla_Y(X(f)s + f\nabla_X s) - [X, Y](f)s - f\nabla_{[X, Y]}s.$$

Using the Leibniz rule, all terms involving derivatives of f cancel, leaving $fR(X, Y)s$. For X : by a direct expansion,

$$R(fX, Y)s = \nabla_{fX} \nabla_Y s - \nabla_Y \nabla_{fX} s - \nabla_{[fX, Y]}s = f\nabla_X \nabla_Y s - \nabla_Y(f\nabla_X s) - \nabla_{f[X, Y] - Y(f)X}s,$$

and $\nabla_Y(f\nabla_X s) = Y(f)\nabla_X s + f\nabla_Y \nabla_X s$ while $\nabla_{f[X, Y] - Y(f)X}s = f\nabla_{[X, Y]}s - Y(f)\nabla_X s$; the two $Y(f)\nabla_X s$ terms cancel, giving $fR(X, Y)s$. ■

In a local frame, the curvature is represented by the *curvature 2-form* $\Omega = (\Omega_j^i)$ given by

$$\Omega = d\omega + \omega \wedge \omega, \quad \text{equivalently} \quad \Omega_j^i = d\omega_j^i + \omega_k^i \wedge \omega_j^k. \quad (2)$$

Under the frame change $e' = eg$ one has

$$\Omega' = g^{-1} \Omega g, \quad (3)$$

the homogeneous transformation law which again exhibits the tensorial character of curvature. Equation (2) is called the (second) *structure equation*.

Proposition 10 (Bianchi identity). The curvature form satisfies

$$d\Omega = \Omega \wedge \omega - \omega \wedge \Omega,$$

or, in the invariant form obtained by taking the covariant exterior derivative d^∇ ,

$$d^\nabla \Omega = 0.$$

Proof. Differentiating (2),

$$d\Omega = d(d\omega) + d(\omega \wedge \omega) = (d\omega) \wedge \omega - \omega \wedge d\omega.$$

Substituting $d\omega = \Omega - \omega \wedge \omega$,

$$d\Omega = (\Omega - \omega \wedge \omega) \wedge \omega - \omega \wedge (\Omega - \omega \wedge \omega) = \Omega \wedge \omega - \omega \wedge \Omega,$$

since $\omega \wedge \omega \wedge \omega - \omega \wedge \omega \wedge \omega = 0$. The covariant exterior derivative of an $\text{End } E$ -valued k -form α is $d^\nabla \alpha = d\alpha + \omega \wedge \alpha - (-1)^k \alpha \wedge \omega$; for $k = 2$ this is $d\Omega + \omega \wedge \Omega - \Omega \wedge \omega = 0$. ■

Remark 11 (Curvature measures non-flatness). A connection with $R \equiv 0$ is *flat*; then parallel transport is locally independent of the path and E admits a local frame of ∇ -parallel sections (the transition functions are then locally constant). Flat connections exist on a bundle exactly when its structure group reduces to a discrete subgroup; for bundles over simply connected bases this forces triviality. The curvature is thus the obstruction to the existence of a parallel frame.

1.4 Parallel transport and holonomy

Let $\gamma : [0, 1] \rightarrow M$ be a smooth path. A section s along γ is *parallel* if $\nabla_{\dot{\gamma}(t)} s(t) = 0$. By the existence and uniqueness theorem for linear ODEs, there is a unique parallel section with prescribed initial value, and we obtain a linear isomorphism

$$P_\gamma : E_{\gamma(0)} \xrightarrow{\cong} E_{\gamma(1)},$$

the *parallel transport* along γ . Parallel transport is independent of reparametrisation, satisfies $P_{\gamma \cdot \delta} = P_\delta \circ P_\gamma$, and for a closed loop based at p yields an automorphism of E_p ; the group of all such automorphisms is the *holonomy group* $\text{Hol}_p(\nabla) \subseteq \text{GL}(E_p)$. By the Ambrose–Singer theorem, its Lie algebra is generated by the values of the curvature at points reachable from p ; informally, curvature is the infinitesimal parallel transport around small loops. This circle of ideas is made precise and globalised in Chapter 2.

Remark 12 (*Restricted and full holonomy*). For a connection ∇ on E , the *holonomy group* $\text{Hol}_p(\nabla)$ consists of parallel transports around all loops based at p , and the *restricted holonomy group* $\text{Hol}_p^0(\nabla)$ of those around contractible loops; Hol_p^0 is the identity component of Hol_p , and $\text{Hol}_p/\text{Hol}_p^0$ is a quotient of $\pi_1(M)$. A *reduction of the holonomy* is a subbundle of E preserved by parallel transport; by the holonomy principle, reductions of E correspond bijectively to subgroups of the structure group containing the holonomy group $\text{Hol}_p(\nabla)$. This is the mechanism by which special geometry (Kähler, hyperkähler, G_2 , $\text{Spin}(7)$) is detected: each is a reduction of the holonomy group of the Levi-Civita connection, and the corresponding characteristic classes (Chern, Euler, Pontryagin) are constrained accordingly.

Remark 13 (*Curvature as a 2-form versus as a tensor*). We have defined curvature twice: as a tensor $R(X, Y)$ in Chapter 1 and as a $\mathfrak{g}$ -valued 2-form Ω in Chapter 2. The dictionary is $R(X, Y) = \Omega(X, Y)$ acting on the associated bundle, i.e. for $E = P \times_G V$ and s a section, $R(X, Y)s = \Omega(X, Y) \cdot s$ (the infinitesimal action of $\Omega(X, Y) \in \mathfrak{g}$ on V). Both viewpoints will be used freely; the 2-form version is the one that couples to the Chern–Weil machinery.

1.5 Existence of connections

Theorem 14. Every vector bundle $E \rightarrow M$ over a paracompact base admits a connection.

Proof. Let $\{U_\alpha\}$ be a locally finite open cover trivialising E , with local frames $e_\alpha = (e_{\alpha,1}, \dots, e_{\alpha,r})$, and let $\{\rho_\alpha\}$ be a smooth partition of unity subordinate to $\{U_\alpha\}$. On each U_α define the *trivial connection* by $\nabla_X^\alpha s = \sum_i X(s^i) e_{\alpha,i}$ for $s = \sum_i s^i e_{\alpha,i}$; it is a connection because $X(fs^i) = X(f)s^i + fX(s^i)$ gives the Leibniz rule. Now set

$$\nabla_X s = \sum_\alpha \rho_\alpha \nabla_X^\alpha s.$$

This is a finite sum at each point. C^∞ -linearity in X is clear. The Leibniz rule:

$$\begin{aligned} \nabla_X(fs) &= \sum_\alpha \rho_\alpha \nabla_X^\alpha(fs) = \sum_\alpha \rho_\alpha (X(f)s + f\nabla_X^\alpha s) \\ &= X(f) \left(\sum_\alpha \rho_\alpha \right) s + f \sum_\alpha \rho_\alpha \nabla_X^\alpha s = X(f)s + f\nabla_X s, \end{aligned}$$

using $\sum_\alpha \rho_\alpha = 1$. Hence ∇ is a connection. ■

Remark 15. The proof shows that a connection is a purely local object that can always be patched together; the interesting question is whether it can be chosen with special properties (flat, metric-compatible, torsion-free), which is where the geometry enters.

1.6 Torsion

For the tangent bundle TM there is a natural tensorial invariant of a connection measuring its failure to be symmetric.

Definition 16 (Torsion). The *torsion* of a connection ∇ on TM is

$$T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y].$$

Proposition 17 (Torsion is a tensor). T is C^∞ -bilinear and skew-symmetric, hence a section of $\bigwedge^2 T^*M \otimes TM$.

Proof. It suffices to check C^∞ -linearity in X :

$$\begin{aligned} T(fX, Y) &= \nabla_{fX} Y - \nabla_Y(fX) - [fX, Y] \\ &= f\nabla_X Y - Y(f)X - f\nabla_Y X - f[X, Y] + Y(f)X \\ &= fT(X, Y). \end{aligned}$$

Skew-symmetry $T(X, Y) = -T(Y, X)$ is immediate from the definition. ■

1.7 Metric connections and the Levi-Civita connection

Definition 18 (Metric connection). Let $E \rightarrow M$ be a vector bundle with a (Riemannian or Hermitian) metric $\langle \cdot, \cdot \rangle$. A connection ∇ is *metric-compatible* (or *metric*) if

$$X\langle s, t \rangle = \langle \nabla_X s, t \rangle + \langle s, \nabla_X t \rangle \quad \forall s, t \in \Gamma(E), X \in \mathfrak{X}(M).$$

The fundamental theorem of Riemannian geometry asserts that on the tangent bundle there is a unique connection which is both metric and torsion-free.

Theorem 19 (Levi-Civita connection). Let (M, g) be a Riemannian manifold. There exists a unique connection ∇ on TM which is metric-compatible and torsion-free. It is determined by the *Koszul formula*

$$\begin{aligned} 2g(\nabla_X Y, Z) &= Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) \\ &\quad + g([X, Y], Z) - g([Y, Z], X) + g([Z, X], Y). \end{aligned}$$

Proof. Uniqueness. Suppose ∇ is metric and torsion-free. Apply the metric compatibility three times:

$$\begin{aligned} Xg(Y, Z) &= g(\nabla_X Y, Z) + g(Y, \nabla_X Z), \\ Yg(Z, X) &= g(\nabla_Y Z, X) + g(Z, \nabla_Y X), \\ Zg(X, Y) &= g(\nabla_Z X, Y) + g(X, \nabla_Z Y). \end{aligned}$$

Adding the first two and subtracting the third, and using torsion-freeness $\nabla_X Y - \nabla_Y X = [X, Y]$ (and cyclically), one obtains exactly the Koszul formula. Since g is non-degenerate, this determines $\nabla_X Y$ uniquely.

Existence. Denote the right-hand side of the Koszul formula by $K(X, Y, Z)$. For fixed X, Y , the functional $Z \mapsto K(X, Y, Z)$ is C^∞ -linear in Z : substituting fZ for Z , the terms not containing f are

$$X(f)g(Y, Z) - X(f)g(Z, Y) + Y(f)g(Z, X) - Y(f)g(Z, X),$$

which vanish by the symmetry of g (here we use $[fZ, X] = f[Z, X] - X(f)Z$ and $[Y, fZ] = f[Y, Z] + Y(f)Z$). By non-degeneracy of g there is a unique vector field $\nabla_X Y$ with $2g(\nabla_X Y, Z) = K(X, Y, Z)$ for every Z . It remains to check the connection axioms.

C^∞ -linearity in X . Substituting fX for X , the terms not containing f are

$$Y(f)g(Z, X) - Y(f)g(X, Z) - Z(f)g(X, Y) + Z(f)g(X, Y),$$

again zero by symmetry of g ; hence $K(fX, Y, Z) = fK(X, Y, Z)$, so $\nabla_{fX} Y = f\nabla_X Y$.

Leibniz rule in Y . Substituting fY for Y , the terms not containing f are

$$X(f)g(Y, Z) + X(f)g(Y, Z) - Z(f)g(X, Y) + Z(f)g(Y, X) = 2X(f)g(Y, Z),$$

using $[X, fY] = f[X, Y] + X(f)Y$ and $[fY, Z] = f[Y, Z] - Z(f)Y$ and the symmetry of g . Thus $K(X, fY, Z) = fK(X, Y, Z) + 2X(f)g(Y, Z)$, i.e. $\nabla_X(fY) = f\nabla_X Y + X(f)Y$.

Metric compatibility. A direct sum gives $K(X, Y, Z) + K(X, Z, Y) = 2Xg(Y, Z)$: the mixed terms group as

$$\begin{aligned} & [Yg(Z, X) - Yg(X, Z)] + [Zg(Y, X) - Zg(X, Y)] \\ & + [g([X, Y], Z) + g([Y, X], Z)] \\ & + [-g([Y, Z], X) - g([Z, Y], X)] + [g([Z, X], Y) + g([X, Z], Y)], \end{aligned}$$

each bracket being zero by symmetry of g or antisymmetry of the bracket. Hence $Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$.

Torsion-freeness. Likewise $K(X, Y, Z) - K(Y, X, Z) = 2g([X, Y], Z)$. The scalar-derivative terms cancel pairwise by symmetry of g , while the bracket terms reduce to

$$\begin{aligned} & g([X, Y], Z) - g([Y, X], Z) \\ & + [-g([Y, Z], X) + g([X, Z], Y) + g([Z, X], Y) - g([Z, Y], X)] = 2g([X, Y], Z), \end{aligned}$$

since $[Y, X] = -[X, Y]$ and the four remaining terms cancel in pairs. Therefore $g(\nabla_X Y - \nabla_Y X, Z) = g([X, Y], Z)$ for all Z , i.e. $\nabla_X Y - \nabla_Y X = [X, Y]$.

In a coordinate frame the Christoffel symbols are

$$\Gamma_{ij}^k = \frac{1}{2}g^{k\ell}(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}),$$

the classical formula. ■

Example 20 (The Levi-Civita connection on $\mathbb{R}^n$). For the Euclidean metric, $\Gamma_{ij}^k = 0$ and ∇ is ordinary directional differentiation; this is the unique flat metric torsion-free connection.

Example 21 (The Levi-Civita connection and curvature of S^2). Let $S^2 \subseteq \mathbb{R}^3$ carry the round metric induced from $\mathbb{R}^3$, with the orthonormal frame e_1, e_2 on the open set where φ is a coordinate. The structure equations (with the convention $de_i = \omega_i^j e_j$) give the connection form $\omega_2^1 = -\cos \theta d\varphi$, and the curvature form

$$\Omega_2^1 = d\omega_2^1 + \omega_k^1 \wedge \omega_2^k = \sin \theta d\theta \wedge d\varphi.$$

The Gauss curvature is $K = \frac{\Omega_2^1(e_1, e_2)}{\det g} = 1$, and the curvature 2-form integrates to $\int_{S^2} \Omega_2^1 = 4\pi$. Since $\text{Pf}(\Omega/2\pi) = \Omega_2^1/2\pi$, the Gauss–Bonnet theorem $\int_{S^2} \frac{\Omega_2^1}{2\pi} = 2 = \chi(S^2)$ is recovered; this is the model for the general transgression argument of Chapter 6.

Remark 22 (Three faces of curvature). The single object Ω carries three meanings that will be conflated throughout: (i) the curvature *tensor* $R(X, Y)s$ of Definition (curvature); (ii) the $\text{End } E$ -valued 2-form Ω ; (iii) for a metric connection, the $\mathfrak{so}(n)$ -valued 2-form. The passage (i) $\leftrightarrow$ (ii) is the identification $\Omega^2(M; \text{End } E) \cong \Omega^2(M) \otimes \Gamma(\text{End } E)$, and (ii) $\leftrightarrow$ (iii) is the metric, which identifies skew-adjoint endomorphisms with $\mathfrak{so}(n)$.

2 Principal Bundles and Connections

The passage from vector bundles to principal bundles is the key structural step of the theory: it separates the “linear algebra” of a bundle (the fibre) from the “symmetry” of the bundle (the structure group), and it is the latter that characteristic classes ultimately measure.

2.1 Principal bundles

Definition 1 (Principal bundle). Let G be a Lie group. A *principal G -bundle* is a smooth map $\pi : P \rightarrow M$ together with a smooth right action $P \times G \rightarrow P$, written $(p, g) \mapsto p \cdot g$, such that

- (i) G acts freely and transitively on each fibre $P_m = \pi^{-1}(m)$;
- (ii) π is G -invariant, i.e. $\pi(p \cdot g) = \pi(p)$;
- (iii) P is locally trivial in the G -equivariant sense: every $m \in M$ has a neighbourhood U and a diffeomorphism $\varphi : \pi^{-1}(U) \rightarrow U \times G$ of the form $\varphi(p) = (\pi(p), \phi(p))$ with $\phi(p \cdot g) = \phi(p)g$.

Equivalently, in symbols, P is a principal G -bundle iff

$$\pi(p \cdot g) = \pi(p), \quad (g \mapsto p \cdot g) : G \xrightarrow{\cong} P_{\pi(p)} \quad (\forall p \in P),$$

and, for every $m \in M$, there are $U \ni m$ and a diffeomorphism $\varphi : \pi^{-1}(U) \xrightarrow{\cong} U \times G$ such that $\varphi(p \cdot g) = \varphi(p) \cdot g$.

Thus each fibre P_m is a copy of G , but without a preferred identity element: a principal bundle is “a Lie group with no chosen origin, varying smoothly over M ”. The fundamental example is the following.

Example 2 (Frame bundle). Let $E \rightarrow M$ be a rank- r vector bundle over $\mathbb{F}$. The *frame bundle* $P_{\text{Fr}}(E)$ has fibre over m equal to the set of all $\mathbb{F}$ -bases (frames) of E_m ; the group $\text{GL}(r; \mathbb{F})$ acts on the right by change of basis, freely and transitively. Equivalently, a frame is a linear isomorphism $\mathbb{F}^r \xrightarrow{\cong} E_m$. Every vector bundle is thus equivalent to a principal $\text{GL}(r; \mathbb{F})$ -bundle, and conversely $E = P \times_G \mathbb{F}^r$ with $G = \text{GL}(r; \mathbb{F})$.

Example 3 (Homogeneous spaces). If $G \rightarrow G/H$ is a homogeneous space of a closed subgroup H , then $G \rightarrow G/H$ is a principal H -bundle. Instances are $\text{SO}(n+1) \rightarrow S^n = \text{SO}(n+1)/\text{SO}(n)$ and $\text{SU}(n+1) \rightarrow \mathbb{C}P^n = \text{SU}(n+1)/\text{U}(n)$.

Definition 4 (Associated bundle). Let $\rho : G \rightarrow \mathrm{GL}(V)$ be a representation. The *associated bundle* is $P \times_{\rho} V := P \times V / \sim$, where $(p, v) \sim (p \cdot g, \rho(g)^{-1}v)$. It is a vector bundle with fibre V . A G -equivariant map between two associated bundles corresponds functorially to a G -equivariant map of the fibres.

Taking $V = \mathbb{F}^r$ and ρ the standard representation recovers the vector bundle whose frame bundle is P . Taking $V = \mathfrak{g}$ and $\rho = \mathrm{Ad}$ gives the *adjoint bundle* $\mathrm{Ad}(P) = P \times_{\mathrm{Ad}} \mathfrak{g}$, whose sections are *infinitesimal gauge transformations*.

2.2 Connections on principal bundles

A connection on a principal bundle is a G -invariant choice of horizontal directions. There are three equivalent formulations.

Definition 5 (Connection: horizontal distribution). A *connection* on $P \rightarrow M$ is a smooth subbundle $H \subseteq TP$ such that

- (i) $T_p P = H_p \oplus V_p$, where $V_p = \ker d\pi_p$ is the vertical subspace;
- (ii) H is G -invariant: $H_{p \cdot g} = (R_g)_* H_p$.

Each $A \in \mathfrak{g}$ determines a *fundamental vertical vector field* $A^{\#}$ on P by $A^{\#}_p = \frac{d}{dt} \big|_{t=0} p \cdot \exp(tA)$. The map $\mathfrak{g} \rightarrow V_p$, $A \mapsto A^{\#}_p$, is a linear isomorphism for every p , so a connection may equivalently be specified by a $\mathfrak{g}$ -valued 1-form annihilating H .

Definition 6 (Connection: connection 1-form). A *connection form* on P is a $\mathfrak{g}$ -valued 1-form $\omega \in \Omega^1(P; \mathfrak{g})$ satisfying

- (i) $\omega(A^{\#}) = A$ for all $A \in \mathfrak{g}$;
- (ii) $R_g^* \omega = \mathrm{Ad}_{g^{-1}} \omega$, i.e. $(R_g^* \omega)_p(v) = \mathrm{Ad}_{g^{-1}} \omega_p(v)$.

The horizontal subspace is then $H_p = \ker \omega_p$, and the two definitions are equivalent. A third, equivalent datum is a G -equivariant horizontal lifting of tangent vectors of M , but we shall not need it. Connections always exist on a principal bundle over a paracompact base (patch local ones by a partition of unity); the space of connections is an affine space modelled on the space of horizontal, Ad -equivariant $\mathfrak{g}$ -valued 1-forms.

2.3 Curvature of a principal connection

Definition 7 (Curvature form). The *curvature form* of a connection ω is the $\mathfrak{g}$ -valued 2-form

$$\Omega = d\omega + \frac{1}{2}[\omega, \omega] \in \Omega^2(P; \mathfrak{g}), \quad (4)$$

where $[\omega, \omega](X, Y) = [\omega(X), \omega(Y)] - [\omega(Y), \omega(X)]$.

Proposition 8 (Properties of the curvature). The curvature form Ω satisfies:

- (i) Ω is horizontal: $\Omega(v, w) = 0$ whenever v or w is vertical;
- (ii) $R_g^* \Omega = \mathrm{Ad}_{g^{-1}} \Omega$ (Ad -equivariance);
- (iii) the *Bianchi identity*: $d\Omega + [\omega, \Omega] = 0$, where $[\omega, \Omega](X, Y, Z)$ is the cyclic sum of

$$[\omega(X), \Omega(Y, Z)].$$

Proof. (i) Let $X = A^\#$ be a fundamental vertical field. For any vector field Y , Cartan's formula gives $d\omega(A^\#, Y) = A^\# \omega(Y) - Y \omega(A^\#) - \omega([A^\#, Y])$. Now $\omega(A^\#) = A$ is a constant element of $\mathfrak{g}$, so $Y \omega(A^\#) = 0$. On the other hand, the Ad-equivariance $R_g^* \omega = \text{Ad}_{g^{-1}} \omega$ differentiates (along $g = \exp tA$) to the infinitesimal relation

$$\mathcal{L}_{A^\#} \omega = -[A, \omega],$$

where $\mathcal{L}$ is the Lie derivative and $[A, \omega]$ denotes the $\mathfrak{g}$ -valued form $Y \mapsto [A, \omega(Y)]$. Since $\mathcal{L}_{A^\#} \omega = d\iota_{A^\#} \omega + \iota_{A^\#} d\omega = dA + \iota_{A^\#} d\omega = \iota_{A^\#} d\omega$, we obtain $d\omega(A^\#, Y) = -[A, \omega(Y)]$. Therefore

$$\Omega(A^\#, Y) = d\omega(A^\#, Y) + [\omega(A^\#), \omega(Y)] = -[A, \omega(Y)] + [A, \omega(Y)] = 0.$$

So Ω vanishes whenever one argument is vertical, proving (i).

(ii) Since $R_g^* d\omega = d(R_g^* \omega)$ and $R_g^* [\omega, \omega] = [R_g^* \omega, R_g^* \omega]$, the equivariance of ω gives

$$R_g^* \Omega = d(\text{Ad}_{g^{-1}} \omega) + \frac{1}{2} [\text{Ad}_{g^{-1}} \omega, \text{Ad}_{g^{-1}} \omega] = \text{Ad}_{g^{-1}} (d\omega + \frac{1}{2} [\omega, \omega]) = \text{Ad}_{g^{-1}} \Omega.$$

(iii) Applying d to (4) and using $d^2 = 0$,

$$d\Omega = \frac{1}{2} d[\omega, \omega] = \frac{1}{2} ([d\omega, \omega] - [\omega, d\omega]) = [d\omega, \omega],$$

the last step because for a $\mathfrak{g}$ -valued 2-form α and a 1-form β one has $[\alpha, \beta] = -[\beta, \alpha]$. Substituting $d\omega = \Omega - \frac{1}{2} [\omega, \omega]$,

$$d\Omega = [\Omega, \omega] - \frac{1}{2} [[\omega, \omega], \omega] = [\Omega, \omega],$$

since $[[\omega, \omega], \omega] = 0$ by the Jacobi identity in $\mathfrak{g}$. Thus $d\Omega + [\omega, \Omega] = 0$. ■

2.4 Local description of a principal connection

Over a trivialising cover $\{U_\alpha\}$ with local sections $s_\alpha : U_\alpha \rightarrow P$, a principal connection form pulls back to local $\mathfrak{g}$ -valued 1-forms

$$A_\alpha = s_\alpha^* \omega \in \Omega^1(U_\alpha; \mathfrak{g}).$$

If $s_\beta = s_\alpha \cdot g_{\alpha\beta}$ with transition functions $g_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow G$, then the A_α are related by

$$A_\beta = \text{Ad}_{g_{\alpha\beta}^{-1}} A_\alpha + g_{\alpha\beta}^{-1} dg_{\alpha\beta}, \quad (5)$$

the familiar gauge transformation law of a Yang–Mills field. This is the principal-bundle ancestor of (1): the connection is a $\mathfrak{g}$ -valued 1-form on P (a geometric object), and the non-homogeneous term appears only after choosing local sections, i.e. local gauges.

Example 9 (The Hopf bundle). The principal $U(1)$ -bundle $S^{2n+1} \rightarrow \mathbb{C}P^n$ (for $n = 1$ the Hopf fibration $S^3 \rightarrow S^2$) carries the connection form induced by the standard metric; its curvature is $2\pi i$ times the Fubini–Study form, and $\frac{i}{2\pi} \int_{\mathbb{C}P^1} \Omega = -1$ records the first Chern number of $\mathcal{O}(-1)$. This is the curvature-theoretic origin of the normalisation axiom (C3) of the Chern classes.

Example 10 (Covering spaces). A principal G -bundle with G discrete is exactly a (normal) covering space of M with deck group G . A connection is then unique and flat (curvature 0), and parallel transport is the monodromy action; the theory collapses to a representation $\pi_1(M) \rightarrow G$. This explains why curvature-theoretic characteristic classes vanish identically on flat bundles.

2.5 Reduction of structure group

For a principal G -bundle $P \rightarrow M$ and a closed subgroup $H \subseteq G$, a *reduction of the structure group to H* is a principal H -bundle $Q \rightarrow M$ with $P \cong Q \times_H G$. Reductions correspond bijectively to sections of the associated bundle $P/H \rightarrow M$. Important instances:

- an orientation of a rank- n real bundle is a reduction of $O(n)$ to $SO(n)$;
- a Riemannian metric is a reduction of $GL(n)$ to $O(n)$ (orthonormal frames);
- a spin structure is a reduction of $SO(n)$ to $Spin(n)$;
- a Hermitian structure on a rank- n complex bundle is a reduction of $GL(n; \mathbb{C})$ to $U(n)$.

Characteristic classes are precisely the obstructions to such reductions, as we shall see for w_1 (orientation) and w_2 (spin) in Chapter 7.

Example 11 (Hermitian and metric connections). Let E be a complex bundle with a Hermitian metric h . The *unitary connections* — those compatible with h — are exactly the connections induced from a principal $U(n)$ -bundle (the frame bundle of unitary frames), i.e. a reduction of the structure group to $U(n)$. For such a connection the curvature takes values in $\mathfrak{u}(n)$, the space of skew-Hermitian matrices, so that $\frac{i}{2\pi} \text{tr } \Omega$ is a real 2-form; this is the reason the first Chern class is a real class, and the starting point of the Chern–Weil normalisation. The same discussion over $\mathbb{R}$ gives the metric connections of the Levi-Civita theory (Chapter 1), whose curvature takes values in $\mathfrak{so}(n)$ and yields the Euler and Pontryagin classes via the Pfaffian and det.

2.6 From principal connections to vector-bundle connections

A principal connection on P induces connections on every associated bundle, in particular on the underlying vector bundle. Conversely, a connection on a vector bundle E is the same as a connection on its frame bundle.

Proposition 12 (Equivalence). Let $E \rightarrow M$ be a vector bundle and $P = P_{\text{Fr}}(E)$ its frame bundle. There is a bijective correspondence between connections on E and connections on P , under which the local connection 1-form $\omega = (\omega_j^i)$ of Construction 7 is the pullback of the principal connection form via the local frame, and the curvature forms are identified: $\Omega^{\text{PC}} = (\Omega_j^i)$ as in (2).

Proof. We construct the two maps and check they are inverse.

Principal $\Rightarrow$ vector. Given a principal connection $H \subseteq TP$, define a connection ∇ on E as follows. Over a local frame $e = (e_1, \dots, e_r)$, the frame is a local section of P , and we set $\omega = (\omega_j^i) = e^* \omega_{\text{PC}}$ (the pullback of the principal connection form). For a section $s = s^i e_i$ and a vector field X , declare

$$\nabla_X s = (X(s^i) + \omega_j^i(X) s^j) e_i.$$

The C^∞ -linearity in X is immediate. For the Leibniz rule, write $s = \sum s^i e_i$ and compute

$$\nabla_X(fs) = (X(fs^i) + \omega_j^i(X)(fs^j)) e_i = (X(f)s^i + fX(s^i) + f\omega_j^i(X)s^j) e_i = X(f)s + f\nabla_X s,$$

using the $\mathbb{R}$ -linearity and the Leibniz rule of X . This is a connection on E . The curvature of this connection is $d\omega + \omega \wedge \omega$ by (2), which is $e^*(d\omega_{\text{PC}} + \frac{1}{2}[\omega_{\text{PC}}, \omega_{\text{PC}}]) = e^*\Omega_{\text{PC}}$, so the curvatures agree.

Vector $\Rightarrow$ *principal*. Conversely, given ∇ on E , define a horizontal distribution on $P = P_{\text{Fr}}(E)$ as follows. A tangent vector $\dot{e} \in T_e P$ at a frame e is the derivative of a curve of frames $e(t) = (e_1(t), \dots, e_r(t))$; declare $\dot{e}$ horizontal iff each $\dot{e}_i(t)$ is ∇ -parallel, i.e. $\nabla_{\dot{\gamma}} \dot{e}_i = 0$ where γ is the projected curve. This defines an m -dimensional distribution H with $TP = H \oplus V$ (the vertical directions correspond to changing the frame without moving the base), which is $\text{GL}(r)$ -invariant because ∇ is invariant under constant changes of frame. Hence H is a principal connection.

The two constructions are inverse by construction (the local connection form $\omega = (\omega_j^i)$ is recovered both ways), which proves the proposition. $\blacksquare$

Remark 13 (Gauge transformations). An automorphism of the principal bundle P covering id_M is a *gauge transformation*. Locally, over a trivialisation, it is a smooth map $g : U \rightarrow G$; under it the connection and curvature forms transform exactly as in (1) and (3). This is the geometric origin of the non-tensorial transformation law of the connection: ω lives on P , where it *is* tensorial, and only its local pullbacks to M acquire the inhomogeneous term.

Example 14 (The Maurer–Cartan form and the trivial bundle). The group G itself is a principal G -bundle over a point, and its *Maurer–Cartan form* $\theta_{\text{MC}} = g^{-1}dg \in \Omega^1(G; \mathfrak{g})$ is a (left-invariant) connection form. Its structure equation is the Maurer–Cartan equation

$$d\theta_{\text{MC}} + \frac{1}{2}[\theta_{\text{MC}}, \theta_{\text{MC}}] = 0,$$

so the curvature vanishes identically: the Maurer–Cartan connection is flat. For a trivial principal bundle $P = M \times G$, the pullback of θ_{MC} by the projection to G is a flat connection, and every flat connection on a simply connected base is of this form. This example is the prototype of the general phenomenon that flat connections correspond to locally constant transition functions.

3 Chern–Weil Theory

We now reach the heart of the subject: a recipe that turns invariant polynomials on the Lie algebra into de Rham cohomology classes of the base, independent of all choices.

3.1 Invariant polynomials

Definition 1 (Invariant polynomial). Let G be a Lie group with Lie algebra $\mathfrak{g}$. A *G -invariant polynomial* of degree k is a symmetric k -linear map $P : \underbrace{\mathfrak{g} \times \dots \times \mathfrak{g}}_k \rightarrow \mathbb{R}$ which is Ad-invariant:

$$P(\text{Ad}_g X_1, \dots, \text{Ad}_g X_k) = P(X_1, \dots, X_k) \quad (g \in G).$$

Let $I^k(G)$ denote the space of invariant polynomials of degree k , and set $I^*(G) = \bigoplus_k I^k(G)$.

The infinitesimal version of Ad-invariance is: for every $Y \in \mathfrak{g}$,

$$\sum_{i=1}^k P(X_1, \dots, [Y, X_i], \dots, X_k) = 0. \quad (6)$$

The spaces $I^*(G)$ form a graded algebra under the symmetric product, which on the diagonal satisfies $(P \cdot Q)(X) = P(X)Q(X)$; the polarised version is obtained by symmetrising. Here and below we write $P(X) := P(X, \dots, X)$ for the associated homogeneous polynomial.

Example 2 (Invariant polynomials for $\mathrm{GL}(n)$). The fundamental invariants are the coefficients of the characteristic polynomial:

$$\det\left(I + \frac{t}{2\pi i}A\right) = \sum_{k=0}^n \sigma_k(A) t^k, \quad A \in \mathfrak{gl}(n; \mathbb{C}),$$

where σ_k is the k -th elementary symmetric function of the eigenvalues. Their homogeneous polarisations are invariant polynomials. Equivalently, the trace powers $A \mapsto \mathrm{tr}(A^k)$ (whose polarisations are invariant) generate $I^*(\mathrm{GL}(n))$; the two families are related by the Newton identities. For the orthogonal group $\mathrm{SO}(2n)$ there is an invariant of degree n , the *Pfaffian* $\mathrm{Pf} : \mathfrak{so}(2n) \rightarrow \mathbb{R}$, defined below in Chapter 6.

3.2 The Chern–Weil homomorphism

Let $\pi : P \rightarrow M$ be a principal G -bundle with connection ω and curvature $\Omega \in \Omega^2(P; \mathfrak{g})$. For $P \in I^k(G)$ define a $2k$ -form on P by

$$P(\Omega) := P(\underbrace{\Omega, \dots, \Omega}_k).$$

Lemma 3 (Invariance and descent). $P(\Omega)$ is horizontal and G -invariant, hence is the pullback of a unique $2k$ -form on M , which we again denote by $P(\Omega) \in \Omega^{2k}(M)$. Moreover this form is closed: $dP(\Omega) = 0$.

Proof. Horizontality: Ω is horizontal (Proposition 8), and the wedge of horizontal forms is horizontal. G -invariance: by the Ad-equivariance $R_g^*\Omega = \mathrm{Ad}_{g^{-1}}\Omega$ and the Ad-invariance of P ,

$$R_g^*P(\Omega) = P(R_g^*\Omega) = P(\mathrm{Ad}_{g^{-1}}\Omega) = P(\Omega).$$

Hence $P(\Omega)$ is a *basic* form and descends to M . Closedness: by the Bianchi identity $d\Omega = -[\omega, \Omega]$ and the infinitesimal invariance (6),

$$dP(\Omega) = \sum_i P(\Omega, \dots, \underbrace{d\Omega}_i, \dots, \Omega) = - \sum_i P(\Omega, \dots, [\omega, \Omega], \dots, \Omega) = 0.$$

■

Definition 4 (Chern–Weil homomorphism). The map

$$w : I^*(G) \longrightarrow H_{\mathrm{dR}}^{2*}(M), \quad P \longmapsto [P(\Omega)],$$

is called the *Chern–Weil homomorphism* of the bundle P .

3.3 Independence of the connection

Theorem 5 (Chern–Weil). The class $[P(\Omega)] \in H_{\text{dR}}^{2k}(M)$ is independent of the choice of connection; consequently $w : I^*(G) \rightarrow H_{\text{dR}}^{2*}(M)$ is a well-defined algebra homomorphism.

The proof uses the following infinitesimal lemma, which is the algebraic core of the entire theory.

Lemma 6. Let $P \in I^k(G)$ and let $\alpha_1, \dots, \alpha_k$ be $\mathfrak{g}$ -valued forms on P of degrees $d_1, \dots, d_k$. Then

$$dP(\alpha_1, \dots, \alpha_k) = \sum_{i=1}^k (-1)^{d_1 + \dots + d_{i-1}} P(\alpha_1, \dots, d^\nabla \alpha_i, \dots, \alpha_k),$$

where $d^\nabla \alpha_i = d\alpha_i + [\omega, \alpha_i]$ is the covariant exterior derivative of a horizontal equivariant form, and the expression $P(\alpha_1, \dots, \alpha_k)$ is formed by wedging and pairing via P .

Proof. By the ordinary Leibniz rule for d applied to the multilinear form P ,

$$dP(\alpha_1, \dots, \alpha_k) = \sum_{i=1}^k (-1)^{d_1 + \dots + d_{i-1}} P(\alpha_1, \dots, d\alpha_i, \dots, \alpha_k). \quad (7)$$

It remains to replace $d\alpha_i$ by $d^\nabla \alpha_i$. We claim that

$$\sum_{i=1}^k (-1)^{d_1 + \dots + d_{i-1}} P(\alpha_1, \dots, [\omega, \alpha_i], \dots, \alpha_k) = 0. \quad (8)$$

Indeed, evaluate at vectors $X_1, \dots, X_N$ (where $N = 1 + \sum d_i$). Each term is a sum, over permutations of the arguments, of expressions

$$P\left(\alpha_1(\dots), \dots, \underbrace{[\omega(X_r), \alpha_i(\dots)]}_i, \dots, \alpha_k(\dots)\right)$$

with a sign. Fixing all the α_j arguments and the tangent vector X_r at which ω is evaluated, the infinitesimal Ad-invariance (6) with $Y = \omega(X_r)$ gives

$$\sum_{i=1}^k P(\alpha_1(\dots), \dots, [Y, \alpha_i(\dots)], \dots, \alpha_k(\dots)) = 0,$$

because the bracketed entry varies through the k slots while the others are fixed. Summing over all permutations and all r yields (8). Adding (8) to (7) proves the lemma. ■

Proof of Theorem 5. Let ω_0, ω_1 be two connection forms, and set $A = \omega_1 - \omega_0 \in \Omega^1(P; \mathfrak{g})$. Then A is horizontal and Ad-equivariant. Consider the affine family

$$\omega_t = \omega_0 + tA, \quad 0 \leq t \leq 1,$$

with curvature $\Omega_t = d\omega_t + \frac{1}{2}[\omega_t, \omega_t]$. Differentiating in t :

$$\frac{d}{dt}\Omega_t = dA + [\omega_t, A] = d^\nabla A.$$

By the chain rule and multilinearity of P ,

$$\frac{d}{dt}P(\Omega_t) = \sum_{i=1}^k P(\Omega_t, \dots, \underbrace{d^{\nabla_t} A}_i, \dots, \Omega_t) = k dP(A, \Omega_t, \dots, \Omega_t),$$

the second equality being Lemma 6 applied with $\alpha_1 = A$ and $\alpha_2 = \dots = \alpha_k = \Omega_t$ (using $d^{\nabla_t} \Omega_t = 0$, the Bianchi identity), together with the symmetry of P , under which all k summands on the left coincide with $P(d^{\nabla_t} A, \Omega_t, \dots, \Omega_t)$. Integrating from $t = 0$ to $t = 1$,

$$P(\Omega_1) - P(\Omega_0) = d \int_0^1 k P(A, \Omega_t, \dots, \Omega_t) dt.$$

The right-hand side is exact, hence $[P(\Omega_1)] = [P(\Omega_0)]$. The algebra homomorphism property follows from $(P \cdot Q)(\Omega) = P(\Omega) \wedge Q(\Omega)$. $\blacksquare$

Definition 7 (Transgression form). The form

$$TP(\omega_0, \omega_1) = k \int_0^1 P(A, \Omega_t, \dots, \Omega_t) dt$$

appearing above is the *transgression* of P . It satisfies $dTP = P(\Omega_1) - P(\Omega_0)$ and is the seed of the secondary invariants (Chern–Simons forms); we return to it in the proof of the Gauss–Bonnet theorem.

3.4 Examples and the Chern–Simons form

Example 8 (Trace powers and their polarisations). For $\mathfrak{gl}(n)$, the homogeneous polynomial $P_k(A) = \text{tr}(A^k)$ has polarisation

$$P_k(A_1, \dots, A_k) = \frac{1}{k!} \sum_{\sigma \in S_k} \text{tr}(A_{\sigma(1)} \cdots A_{\sigma(k)}),$$

which is the invariant polynomial of degree k corresponding to $\text{tr}(A^k)$. The two families $\{\sigma_k\}$ (elementary symmetric) and $\{P_k = \text{tr}(A^k)\}$ are related by the Newton identities

$$k \sigma_k = \sum_{j=1}^k (-1)^{j-1} P_j \sigma_{k-j},$$

so they generate the same ring $I^*(\text{GL}(n))$.

Example 9 (The first Chern class of a line bundle). For a complex line bundle $L \rightarrow M$ the curvature is a purely imaginary 2-form Ω ($\mathfrak{u}(1)$ -valued), and the only nonzero class is

$$c_1(L) = \left[\frac{i}{2\pi} \Omega \right] \in H^2(M; \mathbb{R}).$$

If L carries a Hermitian metric and a compatible connection, then $\frac{i}{2\pi} \Omega$ is a closed real 2-form; by the $dd\bar{\cdot}$ -lemma on a Kähler manifold it represents the de Rham class of the divisor of a meromorphic section. This is the geometric bridge to the Riemann–Roch setting.

Definition 10 (Chern–Simons form). Let ω_0, ω_1 be two connections on P . The Chern–Simons form associated to $P \in I^k(G)$ is

$$\text{CS}_P(\omega_0, \omega_1) = k \int_0^1 P(\omega_1 - \omega_0, \Omega_t, \dots, \Omega_t) dt \in \Omega^{2k-1}(M),$$

where Ω_t is the curvature of $\omega_t = (1-t)\omega_0 + t\omega_1$. It satisfies the *transgression equation*

$$d \text{CS}_P(\omega_0, \omega_1) = P(\Omega_1) - P(\Omega_0).$$

Remark 11 (Secondary characteristic classes). When $P(\Omega_0) = P(\Omega_1) = 0$ (e.g. for two flat connections, or when the characteristic class vanishes for degree reasons), the Chern–Simons form is closed and defines a class in $H^{2k-1}(M; \mathbb{R})$ modulo the image of the Chern–Weil classes; these *secondary* (or Chern–Simons) characteristic classes are not homotopy invariants of the bundle but depend on the connection, and are the starting point of Chern–Simons gauge theory.

Remark 12 (Invariance under the adjoint action). The requirement that P be Ad-invariant is essential: only Ad-invariant polynomials produce well-defined forms $P(\Omega)$ that descend to M and are independent of the connection. This is the point at which the structure group G , rather than the bundle’s topology alone, enters the Chern–Weil construction.

Example 13 (Chern–Weil for an $\text{SU}(2)$ -bundle). Let $P \rightarrow M$ be a principal $\text{SU}(2)$ -bundle. The invariant polynomials of $\mathfrak{su}(2)$ are generated by $P_2(A) = -\frac{1}{8\pi^2} \text{tr}(A^2)$ (the normalisation being chosen so that the resulting class is integral). The Chern–Weil class is

$$-\frac{1}{8\pi^2} [\text{tr}(\Omega \wedge \Omega)] \in H^4(M; \mathbb{R}),$$

the *second Chern class* $c_2(P)$. For the bundle of orthonormal frames of the underlying complex rank-2 bundle of the tautological quaternionic line bundle over $\mathbb{H}P^1 \cong S^4$ (the so-called one-instanton bundle), the integral $\int_{S^4} c_2$ is the *instanton number* (or topological charge) k , an integer that classifies $\text{SU}(2)$ -bundles over S^4 . This is the gauge-theoretic application of Chern–Weil theory: the topological charge is computed by integrating a curvature density.

3.5 Functoriality

Proposition 14 (Naturality). Let $f : N \rightarrow M$ be smooth and $P \rightarrow M$ a principal G -bundle. The Chern–Weil homomorphism of the pullback $f^*P \rightarrow N$ satisfies

$$w_{f^*P} = f^* \circ w_P : I^*(G) \longrightarrow H_{\text{dR}}^{2*}(N).$$

Proof. A connection ω on P pulls back to a connection $f^*\omega$ on f^*P , with curvature $f^*\Omega$. Hence $P(f^*\Omega) = f^*P(\Omega)$, and the statement follows. ■

Remark 15 (The message of Chern–Weil). Theorem 5 is the precise form of the duality announced in the Introduction: the *topological* data (the bundle P , whose isomorphism class is independent of any connection) is probed by *geometric* data (a connection, hence curvature), and the resulting cohomology class is independent of the choice of probe. Every characteristic class constructed from the curvature below is a special case of this principle.

3.6 Compatibility with bundle operations

The Chern–Weil homomorphism is compatible with the standard operations on bundles, which is what makes the characteristic classes computable.

Proposition 16 (Chern–Weil and direct sums). Let P, Q be principal G -bundles with connections ω_P, ω_Q and curvatures Ω_P, Ω_Q . For the product connection on $P \times_M Q$,

$$P_k(\Omega_{P \times Q}) = P_k(\Omega_P \oplus \Omega_Q),$$

so that $w_{P \times Q}(P_k) = w_P(P_k) + w_Q(P_k)$ for the additive invariants such as $\text{tr}(\Omega^k)$ and the corresponding property for the multiplicative ones.

Proof. The curvature of the product connection is block diagonal $\Omega_{P \times Q} = \text{diag}(\Omega_P, \Omega_Q)$, so evaluating a homogeneous polynomial of degree k gives the sum over the two blocks; the statement for the multiplicative invariants follows from the same block-diagonal form and $\det(I + \text{diag}(A, B)) = \det(I + A) \det(I + B)$. ■

Remark 17 (Chern–Weil and reductions). If the structure group of P reduces from G to $H \subseteq G$, then a connection on the reduced bundle induces one on P , and the curvature takes values in $\mathfrak{h} \subseteq \mathfrak{g}$; the Chern–Weil classes of P are then obtained by restricting the invariant polynomial from G to H and evaluating it on the reduced bundle, i.e. $w_P(P) = w_Q(P|_{\mathfrak{h}})$ for the reduction $Q \subseteq P$. This is the mechanism behind the relations $p_k = (-1)^k c_{2k}(E_{\mathbb{C}})$ and $e^2 = p_n$ of Chapters 5 and 6.

4 Chern Classes

Throughout this chapter $E \rightarrow M$ is a complex vector bundle of rank n , with a connection ∇ and curvature Ω (an $\text{End } E$ -valued 2-form, or equivalently the curvature form of the corresponding principal $\text{U}(n)$ -bundle). The invariant polynomials of Example 2 produce, via Chern–Weil, the Chern classes.

4.1 Definition via curvature

Definition 1 (Total Chern class). The *total Chern class* of E is

$$c(E) = \left[\det \left(I + \frac{i}{2\pi} \Omega \right) \right] = 1 + c_1(E) + \cdots + c_n(E) \in H^{2*}(M; \mathbb{R}),$$

where $c_k(E) \in H^{2k}(M; \mathbb{R})$ is the k -th *Chern class*:

$$c_k(E) = \left[\sigma_k \left(\frac{i}{2\pi} \Omega \right) \right],$$

with σ_k the k -th elementary symmetric polynomial of the eigenvalues of $\frac{i}{2\pi} \Omega$. In particular

$$c_1(E) = \left[\frac{i}{2\pi} \text{tr } \Omega \right], \quad c_n(E) = \left[\det \left(\frac{i}{2\pi} \Omega \right) \right].$$

By Theorem 5 each $c_k(E)$ is well defined, i.e. independent of the connection. Since the expression $\det(I + \frac{i}{2\pi} \Omega)$ is Ad -invariant on $\mathfrak{gl}(n; \mathbb{C})$ and $\text{U}(n)$ is a maximal compact subgroup of $\text{GL}(n; \mathbb{C})$ with the same invariant polynomials, the definition is consistent.

Remark 2 (Top Chern class and the Euler class). For a complex bundle, $c_n(E)$ is the Euler class of the underlying oriented real bundle $E_{\mathbb{R}}$ (Chapter 6). This is the first indication that the four families of classes are not independent.

4.2 Axiomatic characterisation

Chern classes are determined by four axioms. The following theorem is the central structural result of this chapter.

Theorem 3 (Axioms for Chern classes). There is a unique assignment, to each complex vector bundle $E \rightarrow M$, of classes $c_k(E) \in H^{2k}(M; \mathbb{R})$ ($k \geq 0$, $c_0 = 1$), satisfying:

- (C1) (*Naturality*) $c_k(f^*E) = f^*c_k(E)$ for any smooth $f : N \rightarrow M$;
- (C2) (*Whitney sum*) $c(E \oplus F) = c(E)c(F)$, i.e. $c_k(E \oplus F) = \sum_{i+j=k} c_i(E)c_j(F)$;
- (C3) (*Normalisation*) for the tautological line bundle $L = \mathcal{O}(-1) \rightarrow \mathbb{C}P^1$, $c_1(L) = -[\omega]$, where ω is the Fubini–Study form with $\int_{\mathbb{C}P^1} \omega = 1$;
- (C4) (*Vanishing*) $c_k(E) = 0$ for $k > \text{rk } E$.

Proof (existence). The Chern–Weil construction provides classes satisfying (C1)–(C4). Naturality is Proposition 14. The Whitney sum formula: give E and F connections with curvatures Ω_E, Ω_F ; the direct sum connection has block-diagonal curvature $\Omega_E \oplus \Omega_F$, hence $\det(I + \frac{i}{2\pi}(\Omega_E \oplus \Omega_F)) = \det(I + \frac{i}{2\pi}\Omega_E) \det(I + \frac{i}{2\pi}\Omega_F)$, giving (C2). Vanishing is clear since the rank of $\frac{i}{2\pi}\Omega$ is n . For (C3) see Example 4 below. ■

Proof (uniqueness) – outline. Uniqueness is a consequence of the splitting principle (Theorem 9) and the computation of the cohomology of the classifying space $\text{BU}(n)$ carried out in Chapter 8. We give the argument there. ■

Example 4 (The tautological line bundle on $\mathbb{C}P^1$). The tautological bundle $\mathcal{O}(-1) \rightarrow \mathbb{C}P^1$ has fibre over a line $\ell \subseteq \mathbb{C}^2$ equal to ℓ itself. Its Fubini–Study connection has curvature form $\Omega = 2\pi i \omega$ (for the round metric on S^2), whence $c_1(\mathcal{O}(-1)) = \frac{i}{2\pi}[2\pi i \omega] = -[\omega]$, verifying (C3). The dual $\mathcal{O}(1)$ has $c_1(\mathcal{O}(1)) = [\omega]$, and the Euler class of $T\mathbb{C}P^1 \cong \mathcal{O}(2)$ is $2[\omega] = \chi(\mathbb{C}P^1)[\omega]$, an early instance of Gauss–Bonnet.

4.3 Basic properties

Proposition 5 (Formal properties of Chern classes). For complex bundles E, F over M :

- (i) $c_k(\bar{E}) = (-1)^k c_k(E)$;
- (ii) $c_k(E^*) = (-1)^k c_k(E)$;
- (iii) if L, L' are line bundles, $c_1(L \otimes L') = c_1(L) + c_1(L')$;
- (iv) $c(E \otimes F)$ is determined by the Chern roots (below).

Proof. (i) The conjugate bundle has curvature the complex conjugate $-\Omega^T$, whose eigenvalues are the negatives of those of Ω ; the elementary symmetric functions change by $(-1)^k$. (ii) E^* is isomorphic to $\bar{E}$ (choose a Hermitian metric), so this follows from (i).

(iii) For line bundles the curvature of the tensor product connection is $\Omega_L + \Omega_{L'}$, and $\frac{i}{2\pi} \operatorname{tr}(\Omega_L + \Omega_{L'}) = \frac{i}{2\pi} \operatorname{tr} \Omega_L + \frac{i}{2\pi} \operatorname{tr} \Omega_{L'}$. ■

4.4 First computations

Example 6 (Chern classes of $\mathbb{C}P^n$). Let $x = c_1(\mathcal{O}(1)) = [\omega] \in H^2(\mathbb{C}P^n; \mathbb{R})$ be the hyperplane class. The *Euler sequence*

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}(1)^{\oplus(n+1)} \longrightarrow T\mathbb{C}P^n \longrightarrow 0$$

exhibits the tangent bundle as a quotient of $\mathcal{O}(1)^{\oplus(n+1)}$ by the trivial line bundle. By the Whitney sum formula,

$$c(T\mathbb{C}P^n) = c(\mathcal{O}(1)^{\oplus(n+1)}) = (1 + x)^{n+1},$$

so that

$$c_k(T\mathbb{C}P^n) = \binom{n+1}{k} x^k, \quad c_n(T\mathbb{C}P^n) = (n+1) x^n.$$

Since $\chi(\mathbb{C}P^n) = n+1$ and $\int_{\mathbb{C}P^n} x^n = 1$, this reads $\int_{\mathbb{C}P^n} c_n(T\mathbb{C}P^n) = n+1 = \chi(\mathbb{C}P^n)$, the Gauss–Bonnet theorem for $\mathbb{C}P^n$.

Example 7 (Chern class of a Riemann surface). For a compact Riemann surface Σ of genus g , the tangent bundle is a complex line bundle with $c_1(T\Sigma) = e(T\Sigma) = \chi(\Sigma)[\omega] = (2 - 2g)[\omega]$, and the Gauss–Bonnet theorem $\int_{\Sigma} c_1 = 2 - 2g$ is the classical $\int_{\Sigma} K dA = 2\pi(2 - 2g)$ in complex form.

Example 8 (Chern class of a hypersurface). Let $X \subseteq \mathbb{C}P^{n+1}$ be a smooth hypersurface of degree d . The adjunction formula identifies the normal bundle $N_X = \mathcal{O}(d)|_X$, and the exact sequence

$$0 \longrightarrow TX \longrightarrow T\mathbb{C}P^{n+1}|_X \longrightarrow N_X \longrightarrow 0$$

gives, writing $h = c_1(\mathcal{O}(1)|_X)$,

$$c(X) = (1 + h)^{n+2}(1 + dh)^{-1} = 1 + (n + 2 - d)h + \cdots.$$

In particular $c_1(X) = (n + 2 - d)h$, so X is Fano (resp. Calabi–Yau, resp. of general type) according as $d < n+2$ (resp. $d = n+2$, resp. $d > n+2$). This is the Chern-class formulation of the classification of hypersurfaces by degree, and is the starting point of the Calabi–Yau / Kähler–Einstein story.

4.5 The splitting principle

Many computations reduce to the case of line bundles by the following device.

Theorem 9 (Splitting principle). For every complex vector bundle $E \rightarrow M$ of rank n there is a manifold $F(E) \rightarrow M$, the *flag bundle*, such that

- (i) the pullback f^*E splits as a direct sum of line bundles, $f^*E \cong L_1 \oplus \cdots \oplus L_n$;
- (ii) $f^* : H^*(M) \rightarrow H^*(F(E))$ is injective.

Consequently, an identity of Chern classes holds on M if and only if it holds after replacing E by a sum of line bundles with first Chern classes $x_i = c_1(L_i)$, the *Chern roots* of E .

Proof. We construct $F(E)$ by induction on the rank. Let $\pi : \mathbb{C}P(E) \rightarrow M$ be the projectivisation, whose fibre over p is $\mathbb{C}P(E_p) \cong \mathbb{C}P^{n-1}$. On $\mathbb{C}P(E)$ there is the tautological line bundle $\mathcal{O}_E(-1) \hookrightarrow \pi^*E$ (the fibre over a line $\ell \subseteq E_p$ is ℓ itself), and we have the exact sequence

$$0 \longrightarrow \mathcal{O}_E(-1) \longrightarrow \pi^*E \longrightarrow Q \longrightarrow 0$$

with Q of rank $n-1$, so $\pi^*E = \mathcal{O}_E(-1) \oplus Q$ (the sequence splits after choosing a Hermitian metric).

We claim $\pi^* : H^*(M) \rightarrow H^*(\mathbb{C}P(E))$ is injective. Let $y = c_1(\mathcal{O}_E(1)) \in H^2(\mathbb{C}P(E); \mathbb{R})$, where $\mathcal{O}_E(1) = \mathcal{O}_E(-1)^*$. Restricted to a fibre $\mathbb{C}P^{n-1}$, y is the hyperplane class h , and $1, h, \dots, h^{n-1}$ is a basis of $H^*(\mathbb{C}P^{n-1}; \mathbb{R})$. By the Leray–Hirsch theorem, the classes $1, y, \dots, y^{n-1}$ form a basis of $H^*(\mathbb{C}P(E); \mathbb{R})$ as a module over $H^*(M; \mathbb{R})$; in particular the map π^* (inclusion of the $H^*(M)$ -submodule generated by 1) is injective.

Iterating (projectivise Q , then its quotient, and so on) yields the full flag bundle $F(E) \rightarrow M$ with $f^*E = L_1 \oplus \dots \oplus L_n$ a sum of line bundles and f^* injective. Writing $x_i = c_1(L_i)$, naturality and the Whitney sum formula give

$$f^*c(E) = c(f^*E) = \prod_{i=1}^n (1 + x_i),$$

and since f^* is injective the total Chern class $c(E)$ is determined by the Chern roots, with $c_k(E) = \sigma_k(x_1, \dots, x_n)$. ■

Corollary 10. Every symmetric polynomial in the Chern roots defines a characteristic class. In particular the *Chern character*

$$\text{ch}(E) = [\text{tr exp}(\frac{i}{2\pi}\Omega)] = \sum_{i=1}^n e^{x_i} = n + c_1 + \frac{c_1^2 - 2c_2}{2} + \dots$$

satisfies $\text{ch}(E \oplus F) = \text{ch}(E) + \text{ch}(F)$ and $\text{ch}(E \otimes F) = \text{ch}(E) \text{ch}(F)$. The Chern character is a ring homomorphism from the Grothendieck group $K(M)$ to $H^{2*}(M; \mathbb{Q})$, and is the natural receptacle for the index theorems.

Example 11 (Low-degree terms of the Chern character). Expanding $\text{ch}(E) = \sum_i e^{x_i}$ in the Chern roots,

$$\text{ch}(E) = n + c_1 + \frac{c_1^2 - 2c_2}{2} + \frac{c_1^3 - 3c_1c_2 + 3c_3}{6} + \dots$$

For a line bundle L this is $\text{ch}(L) = e^{c_1(L)} = 1 + c_1 + \frac{1}{2}c_1^2 + \dots$; for a rank-2 bundle E , $\text{ch}(E) = 2 + c_1 + \frac{c_1^2 - 2c_2}{2}$. These low-degree expressions are the ones that appear in the Riemann–Roch and index computations of Chapter 9.

Remark 12 (Integrality of Chern classes). The Chern–Weil construction produces classes in $H^{2k}(M; \mathbb{R})$. In fact the Chern classes lift canonically to integral classes $c_k(E) \in H^{2k}(M; \mathbb{Z})$ (Theorem 3): for example, $c_1(L)$ of a holomorphic line bundle is the Poincaré dual of the divisor of a meromorphic section, hence integral, and $c_n(E) = e(E_{\mathbb{R}})$ is integral by Gauss–Bonnet. The real classes obtained from the curvature are the images of these integral classes under the change of coefficients $H^*(M; \mathbb{Z}) \rightarrow H^*(M; \mathbb{R})$.

4.6 The Chern class of a line bundle and divisors

For a compact complex manifold M , the group $\text{Pic}(M)$ of holomorphic line bundles is isomorphic to $H^1(M; \mathcal{O}^*)$, and the exponential sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O} \xrightarrow{e^{2\pi i \cdot}} \mathcal{O}^* \rightarrow 0$ gives the connecting map $c_1 : \text{Pic}(M) \rightarrow H^2(M; \mathbb{Z})$. This is the *topological* first Chern class, and it agrees with the curvature class $[\frac{i}{2\pi}\Omega]$ above (both represent the same class in $H^2(M; \mathbb{R})$). This identification — divisor line bundle $\mapsto$ first Chern class — is the geometric content of the Chern class in the complex setting, and is the entry point of the Riemann–Roch theorem.

4.7 Chern numbers

For a compact complex manifold M of complex dimension n , the top-degree Chern classes pair with the fundamental class to give the *Chern numbers*

$$\int_M c_{i_1}(TM) \cdots c_{i_r}(TM), \quad \sum i_j = n.$$

Example 13 (Chern numbers of a complex surface). For a compact complex surface S there are exactly two Chern numbers, $c_1^2 = \int_S c_1(TS)^2$ and $c_2 = \int_S c_2(TS) = \chi(S)$ (the latter by Gauss–Bonnet). By the Noether formula of Example 7-type, $\chi(S, \mathcal{O}_S) = \frac{c_1^2 + c_2}{12}$. The Chern numbers are the discrete invariants that appear throughout the classification of complex surfaces, and are the first instances of the general principle that Chern numbers are cobordism invariants.

Remark 14 (Chern numbers and cobordism). By the analogue of the Pontryagin-number theorem, two stably almost-complex manifolds are complex-cobordant iff they have the same Chern numbers. This is the complex counterpart of the statement for Pontryagin numbers in Chapter 5.

4.8 The Chern character and K -theory

The Chern character of Chapter 4.4 is the natural bridge between the additive structure of vector bundles and the multiplicative structure of cohomology.

Definition 15 (Grothendieck group). The *Grothendieck group* $K(M)$ is the free abelian group on isomorphism classes $[E]$ of complex vector bundles over M , modulo the relation $[E \oplus F] = [E] + [F]$. It is a ring under $\otimes$. Equivalently, in symbols,

$$K(M) = \left(\bigoplus_{[E]} \mathbb{Z} \cdot [E] \right) / \langle [E \oplus F] - [E] - [F] \rangle, \quad [E] \cdot [F] := [E \otimes F].$$

Theorem 16 (Chern character is a ring map). The Chern character induces a ring homomorphism

$$\text{ch} : K(M) \longrightarrow H^{2*}(M; \mathbb{Q}),$$

which becomes an isomorphism $K(M) \otimes \mathbb{Q} \cong H^{2*}(M; \mathbb{Q})$ for M a finite CW complex.

Indication. The additivity $\text{ch}(E \oplus F) = \text{ch}(E) + \text{ch}(F)$ and multiplicativity $\text{ch}(E \otimes F) = \text{ch}(E)\text{ch}(F)$ were established in Chapter 4.4, so ch descends to $K(M)$. The surjectivity onto $H^{2*}(M; \mathbb{Q})$ follows from the splitting principle and the fact that the Newton identities

express any cohomology class as a polynomial in Chern characters of exterior powers; the isomorphism is the content of the Atiyah–Hirzebruch spectral sequence, which we do not develop. Symbolically, the two identities

$$\mathrm{ch}(E \oplus F) = \mathrm{ch}(E) + \mathrm{ch}(F), \quad \mathrm{ch}(E \otimes F) = \mathrm{ch}(E) \mathrm{ch}(F)$$

descend to a well-defined ring map $\mathrm{ch}([E] - [F]) = \mathrm{ch}(E) - \mathrm{ch}(F)$. ■

Remark 17 (Why the Chern character matters). The Chern character is the precise sense in which “characteristic classes are complete”: rationally, the stable isomorphism classes of bundles are classified by the Chern character, and the index theorems express the *analytic* index of an operator as the Chern character of its symbol. It is the single most important structural object of the theory, and the reason the index theorem takes the form $\mathrm{ind} = \int \mathrm{ch} \cdot \mathrm{td}$.

5 Pontryagin Classes

Pontryagin classes are the real analogue of Chern classes, obtained by complexifying.

Definition 1 (Pontryagin classes). Let $E \rightarrow M$ be a real vector bundle of rank n and $E_{\mathbb{C}} = E \otimes_{\mathbb{R}} \mathbb{C}$ its complexification. The *total Pontryagin class* is

$$p(E) = 1 + p_1(E) + \cdots + p_{\lfloor n/2 \rfloor}(E) \in H^{4*}(M; \mathbb{R}), \quad p_k(E) = (-1)^k c_{2k}(E_{\mathbb{C}}) \in H^{4k}(M; \mathbb{R}).$$

The sign is chosen so that $p_k(E)$ has a positive generator in familiar examples. The odd Chern classes of $E_{\mathbb{C}}$ vanish (up to 2-torsion) because $E_{\mathbb{C}} \cong \overline{E_{\mathbb{C}}}$, so that $c_{2k+1}(E_{\mathbb{C}}) = -c_{2k+1}(E_{\mathbb{C}})$.

Proposition 2 (Properties). For real bundles E, F :

- (i) (*Naturality*) $p_k(f^*E) = f^*p_k(E)$;
- (ii) (*Whitney sum*) $p(E \oplus F) = p(E)p(F)$ modulo 2-torsion;
- (iii) if E is a complex bundle of rank n , its underlying real bundle $E_{\mathbb{R}}$ has top Pontryagin class $p_n(E_{\mathbb{R}}) = c_n(E)^2$;
- (iv) for a line bundle L , the real rank-2 bundle $L_{\mathbb{R}}$ satisfies $p_1(L_{\mathbb{R}}) = c_1(L)^2$, and $L_{\mathbb{R}}$ is orientable with Euler class $e(L_{\mathbb{R}}) = c_1(L)$.

Proof. (i) and (ii) follow from the corresponding properties of Chern classes. (iii): when E already carries a complex structure, $(E_{\mathbb{R}})_{\mathbb{C}} \cong E \oplus \bar{E}$, so $p_n(E_{\mathbb{R}}) = (-1)^n c_{2n}(E \oplus \bar{E}) = (-1)^n c_n(E) c_n(\bar{E}) = (-1)^n c_n(E) (-1)^n c_n(E) = c_n(E)^2$. (iv) is the case $n = 1$ of (iii), together with the identification $e = c_1$ for a complex line bundle of Chapter 6. ■

Remark 3 (Two sign conventions). The literature is split on the sign: some authors (Milnor–Stasheff) define $p_k = (-1)^k c_{2k}(E_{\mathbb{C}})$, others (Bott–Tu) work with $(-1)^k c_{2k}$. We follow the first, more common convention, in which the Pontryagin class of the complex projective plane satisfies $p_1(\mathbb{C}P^2) = 3[\omega]^2 > 0$. The difference is immaterial over $\mathbb{R}$ for degree reasons but must be tracked when comparing with the Hirzebruch L -genus.

5.1 Computations

Example 4 (Pontryagin classes of $\mathbb{C}P^n$). Since $c(T\mathbb{C}P^n) = (1 + x)^{n+1}$ with $x = c_1(\mathcal{O}(1))$, the complexification has $c((T\mathbb{C}P^n)_{\mathbb{C}}) = c(T\mathbb{C}P^n)c(\overline{T\mathbb{C}P^n}) = (1 + x)^{n+1}(1 - x)^{n+1} = (1 - x^2)^{n+1}$. Therefore

$$p_k(T\mathbb{C}P^n) = (-1)^k c_{2k}((T\mathbb{C}P^n)_{\mathbb{C}}) = (-1)^k \binom{n+1}{k} (-1)^k x^{2k} = \binom{n+1}{k} x^{2k},$$

that is, $p(T\mathbb{C}P^n) = (1 + x^2)^{n+1}$.

Example 5 (Spheres). The tangent bundle of S^n is *stably trivial*: $TS^n \oplus \varepsilon^1 \cong \varepsilon^{n+1}$ (where ε^k is the trivial rank- k bundle), by the outward normal identification $TS^n \oplus \varepsilon^1 \cong S^n \times \mathbb{R}^{n+1}$. Hence all Pontryagin and Stiefel–Whitney classes of TS^n vanish. This is the simplest instance of the general principle that characteristic classes obstruct triviality of the *stable* bundle.

Definition 6 (Pontryagin numbers). For a closed oriented $4k$ -manifold M and a partition $I = (i_1, \dots, i_r)$ of k , the *Pontryagin number* is

$$p_I(M) = \langle p_{i_1}(TM) \cdots p_{i_r}(TM), [M] \rangle.$$

Theorem 7 (Pontryagin numbers and cobordism). The Pontryagin numbers are cobordism invariants; moreover, by the Thom–Hirzebruch theory, the oriented cobordism ring $\Omega_* \otimes \mathbb{Q}$ is detected by the Pontryagin numbers, and its indecomposables in dimension $4k$ are represented by $\mathbb{C}P^{2k}$. Symbolically,

$$M = \partial W \implies p_I(M) = 0 \quad (I \vdash k),$$

and the evaluation map

$$\Omega_{4k} \otimes \mathbb{Q} \longrightarrow \mathbb{Q}^{\#\{I \vdash k\}}, \quad [M] \longmapsto (p_I(M))_{I \vdash k},$$

is injective, with $\{[\mathbb{C}P^{2k}]\}$ a set of indecomposables in degree $4k$.

Indication. That the Pontryagin numbers are cobordism invariants follows from Stokes' theorem and the fact that the Pontryagin classes of the tangent bundle restrict to those of the boundary, together with the vanishing of the normal bundle on a null-cobordism; the full computation of $\Omega_* \otimes \mathbb{Q}$ uses the multiplicative sequence attached to the L -genus. We do not develop cobordism theory here and refer to Milnor–Stasheff. Symbolically, if $M = \partial W$ with $i : M \hookrightarrow W$ the inclusion, then $i^*TW = TM \oplus \varepsilon^1$, so

$$\begin{aligned} p_I(M) &= \int_M p_{i_1}(TM) \cdots p_{i_r}(TM) \\ &= \int_{\partial W} i^*(p_{i_1}(TW) \cdots p_{i_r}(TW)) \\ &= \int_W d(p_{i_1}(TW) \cdots p_{i_r}(TW)) = 0, \end{aligned}$$

the last step by Stokes' theorem and the closedness of the integrand. ■

Example 8 (The signature of $\mathbb{C}P^2$). By the Hirzebruch signature theorem, $\text{sign}(\mathbb{C}P^2) = \int_{\mathbb{C}P^2} L_1(p_1) = \int_{\mathbb{C}P^2} \frac{p_1}{3} = 1$, since $p_1(\mathbb{C}P^2) = 3x^2$ and $\int_{\mathbb{C}P^2} x^2 = 1$. This is the model computation of the index theorems of Chapter 9.

5.2 Multiplicative sequences: the L -genus and the $\hat{A}$ -genus

The Pontryagin classes generate the *multiplicative sequences*, which are the universal ingredients of the index theorems.

Definition 9 (Multiplicative sequence). A *multiplicative sequence* is a sequence of polynomials $K_k(p_1, \dots, p_k)$ with K_k of weighted degree $4k$ (p_i has weight $4i$), satisfying $K(p \oplus q) = K(p)K(q)$ for the total class $K = 1 + K_1 + K_2 + \dots$. A multiplicative sequence is determined by a formal power series $f(x) = 1 + a_1x + a_2x^2 + \dots$: writing $1 + p_1t + p_2t^2 + \dots = \prod_i (1 + x_i^2t)$ (formally, via the splitting principle), one sets

$$K_k = \text{the degree-}4k \text{ part of } \prod_i f(x_i).$$

Equivalently, via the splitting principle write $p(E) = \prod_i (1 + x_i^2)$ with formal roots x_i ; then, for a formal power series $f(z) = 1 + a_1z + a_2z^2 + \dots$ with $f(0) = 1$,

$$K(E) := \prod_i f(x_i), \quad K_k(E) = [\prod_i f(x_i)]_{4k}, \quad K(E \oplus F) = K(E)K(F),$$

where $[\cdot]_{4k}$ extracts the homogeneous component of degree $4k$.

Example 10 (The L -genus and the $\hat{A}$ -genus). Two multiplicative sequences are ubiquitous:

- the L -genus, with $f(x) = \frac{x}{\tanh x}$: $L_1 = \frac{p_1}{3}$, $L_2 = \frac{7p_2 - p_1^2}{45}$, ..., entering the signature theorem;
- the $\hat{A}$ -genus, with $f(x) = \frac{x/2}{\sinh(x/2)}$: $\hat{A}_1 = -\frac{p_1}{24}$, $\hat{A}_2 = \frac{7p_1^2 - 4p_2}{5760}$, ..., entering the index theorem for the Dirac operator.

Remark 11 (Why multiplicative sequences matter). The multiplicativity $K(E \oplus F) = K(E)K(F)$ means that K factors through the K -theory of the bundle, and the Chern character of Chapter 4 is the bridge: $K(E) = \text{ch}(E)$ composed with the map sending a class to the multiplicative sequence. The index theorems assert that the *index* is such a multiplicative invariant, and the L - and $\hat{A}$ -genera are its two most important specialisations.

6 The Euler Class and the Gauss–Bonnet–Chern Theorem

The Euler class is the characteristic class of an *oriented* real vector bundle, and its integrated form on a closed oriented manifold is the Euler characteristic. We first introduce the Pfaffian, the invariant polynomial that represents it.

6.1 The Pfaffian

Definition 1 (Pfaffian). For $A = (a_{ij}) \in \mathfrak{so}(2n)$, the *Pfaffian* is

$$\text{Pf}(A) = \frac{1}{2^n n!} \sum_{\sigma \in S_{2n}} \text{sgn}(\sigma) a_{\sigma(1)\sigma(2)} a_{\sigma(3)\sigma(4)} \cdots a_{\sigma(2n-1)\sigma(2n)}.$$

Equivalently, $\text{Pf}(A)$ is $1/n!$ times the coefficient of $x_1 \wedge \cdots \wedge x_n$ (up to sign) in the expansion of $\omega_A^{\wedge n}$, where $\omega_A = \frac{1}{2} \sum_{i < j} a_{ij} x_i \wedge x_j$. The fundamental identity is

$$\text{Pf}(A)^2 = \det A, \quad (9)$$

which follows by choosing a basis in which A is block diagonal with 2×2 blocks $\begin{pmatrix} 0 & a_j \\ -a_j & 0 \end{pmatrix}$, so that $\text{Pf}(A) = a_1 \cdots a_n$ and $\det A = (a_1 \cdots a_n)^2$. The Pfaffian is $\text{SO}(2n)$ -invariant: for $B \in \text{GL}(2n)$,

$$\text{Pf}(BAB^T) = \det(B) \text{Pf}(A),$$

so it is invariant under the orientation-preserving group $\text{SO}(2n)$ and changes sign under an orientation reversal.

6.2 Definition of the Euler class

Let $E \rightarrow M$ be an oriented real vector bundle of rank $2n$. Choose a metric on E and a metric-compatible connection ∇ ; its curvature takes values in $\mathfrak{so}(E) \cong \mathfrak{so}(2n)$, so $\text{Pf}(\Omega/2\pi)$ is a $2n$ -form on M .

Definition 2 (Euler class). The *Euler class* of the oriented bundle E is

$$e(E) = \left[\text{Pf} \left(\frac{\Omega}{2\pi} \right) \right] \in H^{2n}(M; \mathbb{R}).$$

By Chern–Weil theory $e(E)$ is independent of the metric and connection; because the frame bundle of an oriented bundle is a principal $\text{SO}(2n)$ -bundle, and Pf is the distinguished degree- n invariant polynomial of $\mathfrak{so}(2n)$, this is a special case of Theorem 5.

Proposition 3 (Basic properties of e).

- (i) $e(E \oplus F) = e(E) e(F)$;
- (ii) $e(-E) = -e(E)$, where $-E$ is E with the reversed orientation;
- (iii) for a complex line bundle L , $e(L_{\mathbb{R}}) = c_1(L)$;
- (iv) $e(E)^2 = p_n(E)$ (the top Pontryagin class).

Proof. (i) is the Whitney sum formula for the Pfaffian (the block-diagonal curvature has Pfaffian the product). (ii) follows from $\text{Pf}(BAB^T) = \det(B) \text{Pf}(A)$ with $\det B = -1$. (iii): for a line bundle the curvature is $\Omega \in \mathfrak{so}(2)$, the single 2×2 block, and $\text{Pf}(\Omega/2\pi) = \frac{i}{2\pi} \text{tr } \Omega$ (the two coincide on $\mathfrak{so}(2) \cong \mathfrak{u}(1)$). (iv): by (9), $\text{Pf}(\Omega/2\pi)^2 = \det(\Omega/2\pi) = p_n(E)$'s curvature representative, computed in Proposition 2(iii). ■

6.3 The angular form and the Gauss–Bonnet–Chern theorem

Let M be a closed oriented Riemannian manifold of even dimension n , and let $\pi : SM \rightarrow M$ be its unit sphere bundle, with fibre S^{n-1} . We write Ω for the curvature form of the Levi-Civita connection (identified with a $\mathfrak{so}(n)$ -valued 2-form on M).

Fix a local orthonormal frame $e_1, \dots, e_n$ with dual coframe $\theta^1, \dots, \theta^n$ and connection forms ω_j^i ($de_j = \omega_j^i e_i$). A point of SM over p is a unit vector $v = \sum v^i e_i(p)$, $\sum (v^i)^2 = 1$. Put $Dv^i = dv^i + \omega_j^i v^j$.

Definition 4 (The angular form). Let ∇ denote the pullback of the Levi-Civita connection to $\pi^*TM \rightarrow SM$. The tautological unit section v determines the orthogonal splitting $\pi^*TM = L \oplus L^\perp$ with $L = \mathbb{R}v$. Let ∇^0 be the connection on π^*TM for which v is parallel and which agrees with ∇ on $L^\perp$:

$$\nabla_X^0 v = 0, \quad \nabla_X^0 = \nabla_X \text{ on sections of } L^\perp.$$

The *angular form* of SM is the transgression of the Pfaffian between ∇^0 and ∇ ,

$$\psi = -n \int_0^1 \text{Pf}\left(\frac{A}{2\pi}, \frac{\Omega_t}{2\pi}, \dots, \frac{\Omega_t}{2\pi}\right) dt, \quad A = \nabla - \nabla^0,$$

where $\nabla_t = (1-t)\nabla^0 + t\nabla$, Ω_t is the curvature of ∇_t , and $\text{Pf}(\cdot, \dots, \cdot)$ is the polarisation of the Pfaffian (the n -fold polarisation of the homogeneous polynomial Pf , defined exactly as for trace powers in Example “Trace powers and their polarisations”). For later use we also record the elementary $(n-1)$ -form

$$\psi_0 = \frac{1}{\sigma_n} \sum_{k=1}^n (-1)^{k-1} v^k Dv^1 \wedge \dots \wedge \widehat{Dv^k} \wedge \dots \wedge Dv^n, \quad \sigma_n = \text{vol}(S^{n-1}) = \frac{2\pi^{n/2}}{\Gamma(n/2)},$$

to which ψ restricts on every fibre (see Lemma 5(ii)).

Lemma 5 (Properties of the angular form). The angular form ψ satisfies:

- (i) $d\psi = -\pi^* \text{Pf}(\Omega/2\pi)$ on SM ;
- (ii) $\int_{S^{n-1}} \psi = 1$, the fibre integral being taken over each sphere $S^{n-1} = \pi^{-1}(p)$.

Proof of (ii). Restrict to a fibre $S^{n-1} = \pi^{-1}(p)$. On the fibre the bundle π^*TM is the constant bundle $T_pM \times S^{n-1}$, and the Levi-Civita connection restricts to the flat connection of the vector space T_pM ; hence on vertical vectors the pullbacks of ω_j^i and Ω_j^i vanish, so $Dv^i = dv^i$ and every curvature factor Ω_t appearing in the polarised Pfaffian of the definition of ψ vanishes on the fibre. Consequently ψ restricts to the term with $n-1$ factors of Dv , namely $\psi_0 = \frac{1}{\sigma_n} \sum_k (-1)^{k-1} v^k dv^1 \wedge \dots \wedge \widehat{dv^k} \wedge \dots \wedge dv^n$, which is $\frac{1}{\sigma_n}$ times the standard solid-angle form on S^{n-1} . The solid-angle form is $\iota_E(dx^1 \wedge \dots \wedge dx^n)$ restricted to the unit sphere, where $E = \sum_i x^i \frac{\partial}{\partial x^i}$ is the Euler (radial) vector field, and its integral over S^{n-1} equals σ_n (a classical computation, e.g. via polar coordinates $dx^1 \wedge \dots \wedge dx^n = r^{n-1} dr \wedge (\text{solid angle})$). Hence $\int_{S^{n-1}} \psi = \int_{S^{n-1}} \psi_0 = 1$. ■

Proof of (i). Apply the transgression identity for the Chern–Simons form (see the definition of the *transgression form* in Chapter 3) to the Pfaffian $P = \text{Pf}(\cdot/2\pi)$ and the two connections ∇^0 and ∇ on $\pi^*TM \rightarrow SM$. Writing $A = \nabla - \nabla^0$, $\nabla_t = (1-t)\nabla^0 + t\nabla$ and Ω_t for the curvature of ∇_t , we have

$$d\left(n \int_0^1 \text{Pf}\left(\frac{A}{2\pi}, \frac{\Omega_t}{2\pi}, \dots, \frac{\Omega_t}{2\pi}\right) dt\right) = \text{Pf}\left(\frac{\Omega}{2\pi}\right) - \text{Pf}\left(\frac{\Omega^0}{2\pi}\right),$$

where Ω, Ω^0 are the curvatures of ∇, ∇^0 . Since $\psi = -n \int_0^1 \text{Pf}(A/2\pi, \Omega_t/2\pi, \dots) dt$ by definition,

$$d\psi = -\text{Pf}\left(\frac{\Omega}{2\pi}\right) + \text{Pf}\left(\frac{\Omega^0}{2\pi}\right).$$

It remains to see that $\text{Pf}(\Omega^0/2\pi) = 0$. Because v is ∇^0 -parallel, for any vector fields X, Y on SM the defining formula of the curvature gives

$$\Omega^0(X, Y)v = \nabla_X^0 \nabla_Y^0 v - \nabla_Y^0 \nabla_X^0 v - \nabla_{[X, Y]}^0 v = 0.$$

Thus the $\mathfrak{so}(n)$ -valued 2-form Ω^0 takes values in the stabiliser of v , that is, in $\mathfrak{so}(v^\perp) \cong \mathfrak{so}(n-1)$ embedded in $\mathfrak{so}(n)$. Any skew endomorphism of $\mathbb{R}^n$ vanishing on the line $L = \mathbb{R}v$ is singular, so its Pfaffian is zero (for a skew matrix $\det = \text{Pf}^2$); therefore $\text{Pf}(\Omega^0/2\pi) = 0$. Hence

$$d\psi = -\text{Pf}\left(\frac{\Omega}{2\pi}\right) = -\pi^* \text{Pf}\left(\frac{\Omega}{2\pi}\right),$$

where the last equality holds because Ω is the pullback of the Levi-Civita curvature of M . $\blacksquare$

We shall also need the following classical result, which computes the Euler characteristic as a sum of local indices.

Definition 6 (Index of an isolated zero). Let X be a vector field with an isolated zero at p . Choose a small ball B around p with no other zero. The *index* $\nu(p)$ of X at p is the degree of the map

$$X/|X| : \partial B \cong S^{n-1} \longrightarrow S^{n-1},$$

the map sending a boundary point to the unit vector $X_q/|X_q|$.

Theorem 7 (Poincaré–Hopf). Let M be a closed oriented manifold and X a vector field with only isolated zeros $p_1, \dots, p_k$. Then

$$\sum_{i=1}^k \nu(p_i) = \chi(M),$$

independently of X .

Indication. Choose a triangulation of M and perturb X so that its zeros are in the interiors of top-dimensional simplices and the flow is transverse to the codimension-one faces. One then shows (by an explicit count of the attaching maps) that the cellular Euler characteristic equals the sum of the indices; the computation is the prototype of the Morse-theoretic argument that $\chi(M) = \sum (-1)^i c_i$ where c_i is the number of critical points of index i of a Morse function. The classical reference is Milnor's *Topology from the Differentiable Viewpoint*. The core identity is

$$\chi(M) = \sum_{i=0}^n (-1)^i \# \{i\text{-cells}\} = \sum_{j=1}^k \nu(p_j),$$

the first equality being the cellular Euler characteristic and the second the count of indices along the flow of X . $\blacksquare$

Theorem 8 (Gauss–Bonnet–Chern). Let M be a closed oriented Riemannian mani-

fold of even dimension n . Then

$$\chi(M) = \int_M \text{Pf}\left(\frac{\Omega}{2\pi}\right) = \int_M e(TM),$$

where $\chi(M) = \sum_i (-1)^i \dim H^i(M; \mathbb{R})$ is the Euler characteristic.

Proof. Choose a vector field X with isolated zeros $p_1, \dots, p_k$ (such fields always exist), and let $\nu(p_i)$ be the index of X at p_i ; by the Poincaré–Hopf theorem $\sum_i \nu(p_i) = \chi(M)$. Remove small open ε -balls $B_\varepsilon(p_i)$ and set $M_\varepsilon = M \setminus \bigcup_i B_\varepsilon(p_i)$.

On M_ε the section X is nowhere zero, hence defines a section $s = X/|X| : M_\varepsilon \rightarrow SM$ of the unit sphere bundle, so that $\pi \circ s = \text{id}_{M_\varepsilon}$. Therefore

$$\text{Pf}\left(\frac{\Omega}{2\pi}\right) \Big|_{M_\varepsilon} = s^* \pi^* \text{Pf}\left(\frac{\Omega}{2\pi}\right) = s^*(-d\psi) = -d(s^*\psi)$$

by Lemma 5(i). Integrating and applying Stokes' theorem,

$$\int_{M_\varepsilon} \text{Pf}\left(\frac{\Omega}{2\pi}\right) = - \int_{M_\varepsilon} d(s^*\psi) = - \int_{\partial M_\varepsilon} s^*\psi = \sum_{i=1}^k \int_{\partial B_\varepsilon(p_i)} s^*\psi. \quad (10)$$

It remains to evaluate each boundary integral. On $\partial B_\varepsilon(p_i)$, the map $s = X/|X|$ restricts to a map $S^{n-1} \rightarrow S^{n-1}$ of degree equal to the index $\nu(p_i)$; moreover ψ restricts on each fibre to the normalised solid-angle form of Lemma 5(ii). Hence

$$\int_{\partial B_\varepsilon(p_i)} s^*\psi \xrightarrow{\varepsilon \rightarrow 0} \nu(p_i) \int_{S^{n-1}} \psi = \nu(p_i),$$

the boundary-orientation sign having already been absorbed in the passage from ∂M_ε to $\partial B_\varepsilon(p_i)$ in (10). Letting $\varepsilon \rightarrow 0$ in (10) gives

$$\int_M \text{Pf}\left(\frac{\Omega}{2\pi}\right) = \sum_i \nu(p_i) = \chi(M),$$

which is the theorem. ■

Remark 9 (The Gauss–Bonnet–Chern theorem). The theorem generalises the classical Gauss–Bonnet formula $\int_M K dA = 2\pi\chi(M)$ for a surface, where K is the Gauss curvature and $\text{Pf}(\Omega/2\pi) = \frac{K}{2\pi} dA$. Chern's 1944 achievement was to extend it to all even dimensions; the integrand $\text{Pf}(\Omega/2\pi)$ is a universal polynomial in the curvature, and the result is a striking instance of a *local* geometric density integrating to a *global* topological invariant. The same phenomenon recurs throughout the index theorems: a characteristic class integrated over M equals an analytic index.

Example 10 (The classical Gauss–Bonnet for surfaces). Let (Σ, g) be an oriented Riemannian surface. In an orthonormal frame the curvature form of the Levi-Civita connection is

$$\Omega = \begin{pmatrix} 0 & \omega \\ -\omega & 0 \end{pmatrix}, \quad \omega = K dA,$$

where K is the Gauss curvature and dA the area form. Then $\text{Pf}(\Omega/2\pi) = \omega/2\pi = \frac{K}{2\pi} dA$, and the Gauss–Bonnet theorem of Theorem 8 gives

$$\int_\Sigma K dA = 2\pi\chi(\Sigma).$$

For $\Sigma = S^2$ (round metric, $K \equiv 1$), this is $4\pi = 2\pi \cdot 2$; for a genus- g surface with a metric of constant curvature -1 (Poincaré metric), $-\text{Area} = 2\pi(2 - 2g)$, i.e. $\text{Area} = 4\pi(g - 1)$.

Example 11 (Euler class of the tangent bundle of $\mathbb{C}P^n$). From Example 6, $e(T\mathbb{C}P^n) = c_n(T\mathbb{C}P^n) = (n + 1)x^n$, and $\int_{\mathbb{C}P^n} e = \int (n + 1)x^n = n + 1 = \chi(\mathbb{C}P^n)$, recovering the Euler characteristic via Gauss–Bonnet–Chern.

Example 12 (Euler class of TS^{2n}). For the even-dimensional sphere S^{2n} , the round metric has constant curvature, and the Euler class of the tangent bundle is $e(TS^{2n}) = 2[\text{vol}]$, where $[\text{vol}]$ is the volume form normalised so that $\int_{S^{2n}} \text{vol} = 1$: the Gauss–Bonnet integrand is 2vol . Hence $\int_{S^{2n}} e = 2 = \chi(S^{2n})$. The factor 2 records that the Gauss–Bonnet integrand of a unit sphere is $\frac{1}{\text{vol}(S^{2n})} \text{vol}$ times the Euler characteristic, a constant-curvature specialisation of the general theorem.

Remark 13 (Euler class of an odd-rank bundle). For an oriented bundle of odd rank the Euler class vanishes: $e(E) = -e(E)$ by reversing orientation, and $e(E) \in H^{\text{odd}}(M; \mathbb{R})$ has no 2-torsion. This is why the Euler class is defined only for even-rank bundles and why the Gauss–Bonnet theorem concerns even-dimensional manifolds. The odd-rank analogue is the vanishing of the Euler class of TS^{2n+1} , consistent with $\chi(S^{2n+1}) = 0$.

6.4 The Thom class and the Euler class

There is a second, topological definition of the Euler class which connects it with the Poincaré duality between a manifold and its zero section, and which is the basis for the general Thom isomorphism.

Let $\pi : E \rightarrow M$ be an oriented real vector bundle of rank n , and let $E_0 \subseteq E$ be the complement of the zero section.

Definition 14 (Thom class). A *Thom class* of E is a class $u \in H^n(E, E_0; \mathbb{R})$ whose restriction to each fibre $(E_p, E_p \setminus \{0\})$ is the orientation class; equivalently, u is the unique class with $\pi_*(u) = 1 \in H^0(M; \mathbb{R})$, where π_* is integration along the fibre. Symbolically, u is the unique class with

$$i_p^* u = [\text{or}_p] \in H^n(E_p, E_p \setminus \{0\}; \mathbb{R}) \quad (\forall p \in M), \quad \text{equivalently} \quad \pi_! u = 1 \in H^0(M; \mathbb{R}),$$

where $i_p : (E_p, E_p \setminus \{0\}) \hookrightarrow (E, E_0)$ is the fibre inclusion, $[\text{or}_p]$ the orientation class of the fibre, and $\pi_!$ integration along the fibre.

Theorem 15 (Thom isomorphism). The cup product with u gives an isomorphism

$$H^k(M; \mathbb{R}) \xrightarrow{\cong} H^{k+n}(E, E_0; \mathbb{R}), \quad \alpha \mapsto \pi^* \alpha \wedge u,$$

the *Thom isomorphism*.

Definition 16 (Euler class via the Thom class). The *Euler class* $e(E) \in H^n(M; \mathbb{R})$ is the pullback of the Thom class to the zero section:

$$e(E) = s_0^* u,$$

where $s_0 : M \rightarrow E$ is the zero section.

Proposition 17 (The two definitions agree). For an oriented Riemannian bundle, the Chern–Weil Euler class $[\text{Pf}(\Omega/2\pi)]$ coincides with the Thom-class Euler class.

Indication. Both classes restrict on fibres to the orientation class and both pull back to the zero section to the class whose integral is the Euler characteristic; by the uniqueness part of the Thom isomorphism and the Gauss–Bonnet theorem they agree. Equivalently, the angular form ψ of Lemma 5 is the seed of the Thom form: its radial extension to (E, E_0) is a closed relative form whose fibre integral is 1, and the pullback of the resulting Thom class along the zero section is exactly $\text{Pf}(\Omega/2\pi)$. The two constructions are the same object seen from the top and from the bottom. Symbolically, with ψ the angular form,

$$\pi_![\psi] = 1, \quad s_0^*u = [\text{Pf}(\Omega/2\pi)], \quad \text{hence} \quad e(E) = s_0^*u = [\text{Pf}(\Omega/2\pi)].$$

■

Remark 18 (Why the Euler class is “the” orientation class). The Euler class is the characteristic class of *oriented* bundles because it measures the twisting of the orientation: $e(E) = 0$ for an odd-rank bundle, and $e(E)$ is the obstruction to finding a nowhere-vanishing section (for $E = TM$, the obstruction to a non-vanishing vector field, by Poincaré–Hopf). This is the topological meaning behind the curvature formula.

Proposition 19 (Euler class and sections). Let $E \rightarrow M$ be an oriented rank- n vector bundle with M closed. Then $e(E) \neq 0$ in $H^n(M; \mathbb{R})$ implies that every section of E vanishes somewhere; conversely, a nowhere-vanishing section forces $e(E) = 0$.

Proof. A nowhere-vanishing section s trivialises a rank-1 direct summand: $E \cong E' \oplus \varepsilon^1$, so $e(E) = e(E')e(\varepsilon^1) = 0$ since $e(\varepsilon^1) = 0$ (a trivial line bundle has odd rank, hence $e = 0$). Contrapositively, $e(E) \neq 0$ forbids a nowhere-zero section. For $E = TM$ this is the statement that a closed manifold with $\chi(M) \neq 0$ admits no non-vanishing vector field, recovering the Poincaré–Hopf obstruction from the Euler class. ■

Remark 20 (The Euler class and the fundamental class). For $E = TM$ over a closed oriented n -manifold, the Euler class evaluated on the fundamental class is the Euler characteristic, $\langle e(TM), [M] \rangle = \chi(M)$, by Gauss–Bonnet. This single evaluation packs the entire Poincaré–Hopf theorem into a cohomological statement, and is the paradigm for all the characteristic *numbers* of Chapters 4 and 5.

7 Stiefel–Whitney Classes

Stiefel–Whitney classes are the mod-2 characteristic classes of real vector bundles. Unlike the previous chapters, they cannot be produced by Chern–Weil theory, because de Rham cohomology with real coefficients cannot see 2-torsion; they are instead defined topologically and characterised by axioms.

Definition 1 (Stiefel–Whitney classes). Let $E \rightarrow M$ be a real vector bundle of rank n . Its *Stiefel–Whitney classes* are the classes $w_k(E) \in H^k(M; \mathbb{Z}/2)$, $k \geq 0$ ($w_0 = 1$), uniquely determined by:

(SW1) (*Naturality*) $w_k(f^*E) = f^*w_k(E)$;

(SW2) (*Whitney sum*) $w_k(E \oplus F) = \sum_{i+j=k} w_i(E) w_j(F)$;

- (SW3) (*Normalisation*) for the tautological line bundle $\gamma^1 \rightarrow \mathbb{R}P^\infty$, $w_1(\gamma^1) \neq 0$;
 (SW4) (*Vanishing*) $w_k(E) = 0$ for $k > \text{rk } E$.

Existence and uniqueness follow from the computation of the mod-2 cohomology of the classifying space $\text{BO}(n)$ (Chapter 8): $H^*(\text{BO}(n); \mathbb{Z}/2) \cong \mathbb{Z}/2[w_1, \dots, w_n]$.

Proposition 2 (Geometric meaning of w_1 and w_2).

- (i) E is orientable $\iff w_1(E) = 0$;
- (ii) E admits a spin structure $\iff w_1(E) = w_2(E) = 0$;
- (iii) $w_1(E) = w_1(\det E)$;
- (iv) if E is the underlying real bundle of a complex bundle F , then $w_{2k}(E) \equiv c_k(F) \pmod{2}$ and $w_{2k+1}(E) = 0$.

Indication. (i) Orientability is equivalent to the structure group reducing from $\text{O}(n)$ to $\text{SO}(n)$, which is exactly the vanishing of the determinant obstruction $w_1 = \det$. (ii) is the corresponding statement for the double cover $\text{Spin}(n) \rightarrow \text{SO}(n)$. (iii) is (i) applied to the determinant line bundle. (iv) is the reduction mod 2 of the Chern classes (the complex orientation kills the odd classes). Symbolically,

$$\begin{aligned} w_1(E) = 0 &\iff E \text{ orientable}, & w_1(E) = w_2(E) = 0 &\iff E \text{ spin}, \\ w_1(E) = w_1(\det E), & & w(E_{\mathbb{R}}) \equiv c(F) \pmod{2} & \text{ (i.e. } w_{2k} \equiv c_k, w_{2k+1} = 0). \end{aligned}$$

■

Remark 3 (Relation to the Euler class). For an oriented real bundle E of rank n , the top Stiefel–Whitney class is the reduction mod 2 of the Euler class: $w_n(E) \equiv e(E) \pmod{2}$. For a closed manifold this is the statement that $\chi(M) \equiv \int_M w_n(TM) \pmod{2}$.

Remark 4 (Wu classes). The *Stiefel–Whitney classes* of a closed manifold determine, and are determined by, its mod-2 Steenrod squares via the Wu classes v_k : $w(TM) = \text{Sq}(v)$, where Sq is the total Steenrod square. In particular w_1, w_2 are detected by the squares Sq^1, Sq^2 , which is the basis of the obstruction-theoretic interpretation of w_1 and w_2 .

7.1 Computations

Example 5 (Stiefel–Whitney classes of $\mathbb{R}P^n$). Let $a = w_1(\gamma^1) \in H^1(\mathbb{R}P^n; \mathbb{Z}/2)$ be the generator of the tautological line bundle. There is an isomorphism

$$T\mathbb{R}P^n \oplus \varepsilon^1 \cong \gamma^1 \oplus \dots \oplus \gamma^1 \quad (n+1 \text{ copies}),$$

so by the Whitney sum formula and the vanishing of the classes of a trivial bundle,

$$w(T\mathbb{R}P^n) = (1 + a)^{n+1} \in H^*(\mathbb{R}P^n; \mathbb{Z}/2) \cong \mathbb{Z}/2[a]/(a^{n+1}).$$

In particular $\mathbb{R}P^n$ is orientable iff n is odd (then $w_1 = (n+1)a \equiv 0$), and $\mathbb{R}P^n$ admits a spin structure iff $n \equiv 3 \pmod{4}$.

Example 6 (The Möbius band and orientability). The tautological bundle $\gamma^1 \rightarrow \mathbb{R}P^1 = S^1$ is the (open) Möbius band. Its w_1 is the generator of $H^1(S^1; \mathbb{Z}/2) \cong \mathbb{Z}/2$, and $w_1 \neq 0$ is precisely the non-orientability of the Möbius band. This is the model example for (i) of Proposition 2.

Example 7 (Stiefel–Whitney classes of a surface). For a closed surface Σ , $w_1(\Sigma) = 0$ iff Σ is orientable, and $w_2(\Sigma) \equiv \chi(\Sigma) \pmod{2}$, the value being read in $H^2(\Sigma; \mathbb{Z}/2) \cong \mathbb{Z}/2$; in particular w_2 vanishes on orientable surfaces but not always on non-orientable ones (e.g. $w_2 = 0$ for the Klein bottle). Thus a spin structure exists on Σ iff $w_1 = w_2 = 0$, i.e. iff Σ is orientable; for an orientable surface the Arf invariant of the mod-2 intersection form (a quadratic refinement) then distinguishes among its spin structures.

Remark 8 (Why the Whitney sum formula is exact here). Over $\mathbb{Z}/2$ there is no sign to track, so the Whitney sum formula $w(E \oplus F) = w(E)w(F)$ holds without any 2-torsion caveat, in contrast to the Pontryagin classes of Proposition 2(ii). This is one reason Stiefel–Whitney classes are the most elementary and computable of the four families.

7.2 Existence and uniqueness

The axioms (SW1)–(SW4) determine the Stiefel–Whitney classes uniquely, and the argument is the mod-2 analogue of the proof completed in Chapter 8 for the Chern classes; we spell out the reduction.

Theorem 9. There is a unique assignment of Stiefel–Whitney classes satisfying (SW1)–(SW4).

Proof. Existence. For a real line bundle $L \rightarrow M$, the unit circle bundle $\pi : S(L) \rightarrow M$ is a double cover (a principal $\mathbb{Z}/2$ -bundle), and its isomorphism class is classified by an element $w_1(L) \in H^1(M; \mathbb{Z}/2)$ under the standard identification of $H^1(M; \mathbb{Z}/2)$ with the set of double covers of M (via $\text{Hom}(\pi_1(M), \mathbb{Z}/2)$). Concretely, $w_1(L)$ is the obstruction to a reduction of the structure group of L from $O(1)$ to $SO(1)$, i.e. to the orientability of L . For line bundles the axioms hold as follows. (SW1) is immediate: pulling back L pulls back its circle bundle. (SW3) holds because the circle bundle of the tautological line bundle $\gamma^1 \rightarrow \mathbb{R}P^\infty$ is the non-trivial double cover, so $w_1(\gamma^1) \neq 0$. (SW2) reduces to the identity $w_1(L_1 \otimes L_2) = w_1(L_1) + w_1(L_2)$, which holds because $\det(L_1 \otimes L_2) \cong \det L_1 \otimes \det L_2$ and tensor products of double covers correspond to sums in $H^1(M; \mathbb{Z}/2)$. (SW4) is automatic for a rank-one bundle.

For a rank- n bundle E , choose a splitting map $f : F \rightarrow M$ with f^* injective on $H^*(-; \mathbb{Z}/2)$ and $f^*E \cong L_1 \oplus \cdots \oplus L_n$, and define $w(E) \in H^*(M; \mathbb{Z}/2)$ by

$$f^*w(E) = \prod_{i=1}^n (1 + w_1(L_i)).$$

The right-hand side is a symmetric polynomial in the $w_1(L_i)$; by the fundamental theorem of symmetric polynomials it therefore lies in $f^*H^*(M; \mathbb{Z}/2)$, and by injectivity of f^* it determines $w(E)$ uniquely. Moreover the definition is independent of the choices: a different splitting merely permutes the L_i , leaving the symmetric product unchanged. The axioms now follow. (SW1): naturality is built into the functorial construction—pulling back a splitting of E along $h : N \rightarrow M$ yields a splitting of h^*E , and the resulting products agree. (SW2): if $f^*E' \cong L'_1 \oplus \cdots \oplus L'_{n'}$, then $f^*(E \oplus E')$ splits into the concatenation of the two lists, whence $f^*w(E \oplus E') = \prod_i (1 + w_1(L_i)) \prod_j (1 + w_1(L'_j)) = f^*w(E) f^*w(E')$, and injectivity of f^* gives $w(E \oplus E') = w(E)w(E')$. (SW3) is the line-bundle case. (SW4): the product $\prod_i (1 + w_1(L_i))$ is concentrated in degrees $\leq n$, so $w_k(E) = 0$ for $k > n$.

Uniqueness. Any assignment satisfying (SW1)–(SW4) is determined by its value on the universal bundle $\gamma^n \rightarrow \text{BO}(n)$, by naturality and the classification theorem. On $\text{BO}(n)$

the splitting principle and the Whitney sum formula force $w(\gamma^n) = \prod_i (1 + a_i)$ with a_i the line-bundle classes, and since $H^*(\mathrm{BO}(n); \mathbb{Z}/2) = \mathbb{Z}/2[w_1, \dots, w_n]$ is a polynomial ring (Theorem 3), the axioms determine $w(\gamma^n)$ uniquely. ■

Proposition 10 (Further properties).

- (i) $w(E) = 1$ (i.e. all $w_k = 0$) whenever E is stably trivial;
- (ii) $w_k(E) = 0$ for $k > n$ and $w_n(E) = e(E) \pmod{2}$ for E oriented;
- (iii) for the double cover $\tilde{M} \rightarrow M$ determined by $w_1(TM) \in H^1(M; \mathbb{Z}/2)$, the pullback kills w_1 .

Proof. (i): a trivial summand contributes nothing, so a stably trivial bundle has $w(E) = 1$; the converse fails in general (e.g. the underlying real rank-4 bundle of the tautological quaternionic line bundle over S^4 has $w = 1$ but $p_1 \neq 0$). (ii): the vanishing is axiom (SW4), and $w_n \equiv e \pmod{2}$ follows from the identification of the top class with the Euler class mod 2. (iii) is the definition of the orientation double cover. ■

8 Classifying Spaces and Universal Classes

This chapter supplies the “global” framework that unifies the preceding constructions and proves the uniqueness parts of the axiomatic characterisations. The idea, due to Steenrod and Milnor, is that every vector bundle is pulled back from a single *universal* bundle over a *classifying space*, and characteristic classes are simply elements of the cohomology of that space.

8.1 Grassmannians and the universal bundle

Definition 1 (Grassmannian and tautological bundle). Let $\mathrm{Gr}_k(\mathbb{F}^N)$ be the Grassmannian of k -dimensional subspaces of $\mathbb{F}^N$ ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$). The *tautological bundle* $\gamma_N^k \rightarrow \mathrm{Gr}_k(\mathbb{F}^N)$ has fibre over a subspace W equal to W itself. Passing to the direct limit $\mathbb{F}^\infty = \bigcup_N \mathbb{F}^N$ gives $\mathrm{Gr}_k(\mathbb{F}^\infty)$ and the universal bundle $\gamma^k \rightarrow \mathrm{Gr}_k(\mathbb{F}^\infty)$. Symbolically,

$$\mathrm{Gr}_k(\mathbb{F}^N) = \{ W \subseteq \mathbb{F}^N : \dim_{\mathbb{F}} W = k \}, \quad \gamma_N^k = \{ (W, v) : W \in \mathrm{Gr}_k(\mathbb{F}^N), v \in W \},$$

with projection $\gamma_N^k \rightarrow \mathrm{Gr}_k(\mathbb{F}^N)$, $(W, v) \mapsto W$.

The Grassmannian is a smooth compact manifold of dimension $k(N - k)$ (respectively $2k(N - k)$ over $\mathbb{C}$), and is the prototype of a *classifying space*.

Theorem 2 (Classification of vector bundles). Let M be paracompact. Then pull-back along the classifying map gives a bijection

$$\begin{aligned} [M, \mathrm{Gr}_k(\mathbb{F}^\infty)] &\xrightarrow{\cong} \{\text{isomorphism classes of rank-}k \text{ } \mathbb{F}\text{-bundles over } M\}, \\ [f] &\longmapsto f^* \gamma^k. \end{aligned}$$

Existence. Embed E as a subbundle of a trivial bundle: for N sufficiently large there is a bundle monomorphism $E \hookrightarrow M \times \mathbb{F}^N$ (choose finitely many generating sections over a finite

cover, by paracompactness). Then each fibre E_p becomes a k -subspace of $\mathbb{F}^N$, and the map $f(p) = [E_p] \in \text{Gr}_k(\mathbb{F}^N)$ satisfies $f^*\gamma^k \cong E$. For uniqueness, two classifying maps f_0, f_1 with $f_0^*\gamma^k \cong f_1^*\gamma^k$ are related by a bundle isomorphism over $M \times [0, 1]$ whose classifying map is a homotopy $f_0 \simeq f_1$; this is the standard homotopy-lifting argument, which we omit. ■

Writing $\text{BO}(k) = \text{Gr}_k(\mathbb{R}^\infty)$, $\text{BU}(k) = \text{Gr}_k(\mathbb{C}^\infty)$, and $\text{BSO}(k)$ the Grassmannian of *oriented* k -planes (the double cover of $\text{BO}(k)$), the theorem identifies principal G -bundle isomorphism classes with $[M, BG]$. The spaces BG are the *classifying spaces*; they are the base of a universal principal bundle $EG \rightarrow BG$ with EG contractible (for $G = \text{O}(k)$ one takes the Stiefel manifold $\text{St}_k(\mathbb{R}^\infty)$, the space of orthonormal k -frames, which is contractible).

8.2 Universal characteristic classes

Because characteristic classes are natural (Proposition 14), the class of the *universal* bundle determines the class of every bundle via pullback along the classifying map. The computation of the cohomology of the classifying spaces is therefore the central calculation of the theory.

Theorem 3 (Cohomology of classifying spaces).

- (i) $H^*(\text{BU}(n); \mathbb{Z}) = \mathbb{Z}[c_1, \dots, c_n]$, c_i the i -th Chern class of γ^n ;
- (ii) $H^*(\text{BO}(n); \mathbb{Z}/2) = \mathbb{Z}/2[w_1, \dots, w_n]$, w_i the i -th Stiefel–Whitney class of γ^n ;
- (iii) $H^*(\text{BSO}(2n); \mathbb{R}) = \mathbb{R}[p_1, \dots, p_{n-1}, e]$, p_i, e the Pontryagin and Euler classes, $e^2 = p_n$.

Indication. For (i) one uses the Schubert cell decomposition of $\text{Gr}_n(\mathbb{C}^N)$: the cells are indexed by partitions $\lambda \subseteq (n \times (N - n))$, of real codimension $2|\lambda|$, giving a CW structure with cells only in even dimensions, hence $H^*(\text{BU}(n); \mathbb{Z})$ is free and equals $\mathbb{Z}[c_1, \dots, c_n]$ (the Chern classes of γ^n are the classes of the codimension- $2i$ Schubert cycles). (ii) is the mod-2 version, with one cell per partition in every dimension. (iii) follows from (i) and (ii) via the relation between the three groups. Symbolically, the even-dimensional cells give

$$H^{2k}(\text{Gr}_n(\mathbb{C}^N); \mathbb{Z}) \cong \mathbb{Z}\langle [\bar{e}_\lambda] : |\lambda| = k \rangle, \quad c_i = [\bar{e}_{(1^i)}],$$

so that $H^*(\text{BU}(n); \mathbb{Z}) = \mathbb{Z}[c_1, \dots, c_n]$ with $\deg c_i = 2i$. ■

Definition 4 (Universal characteristic class). A *universal characteristic class* is an element of $H^*(BG)$. The characteristic class of a G -bundle $P \rightarrow M$ with classifying map $f : M \rightarrow BG$ is f^* of the universal class. Theorem 3 says exactly that the Chern, Stiefel–Whitney, Pontryagin and Euler classes are a complete set of universal characteristic classes. Symbolically, a universal class $u \in H^*(BG)$ assigns to each G -bundle P the class

$$c_u(P) := f^*u \in H^*(M), \quad f : M \rightarrow BG \text{ a classifying map for } P,$$

and the assignment $u \mapsto c_u$ is a bijection onto the natural characteristic classes.

Proof of uniqueness in Theorem 3 – completed. Any assignment $E \mapsto c(E)$ satisfying the axioms is determined by its value on $\gamma^n \rightarrow \text{BU}(n)$ by naturality (Theorem 2), and the

Whitney sum formula plus the splitting principle force $c(\gamma^n) = \prod_i (1 + x_i)$ with $x_i = c_1$ of the line bundles in a splitting. Since $H^*(\mathrm{BU}(n)) = \mathbb{Z}[c_1, \dots, c_n]$ is a polynomial ring with no relations, the axioms determine $c(\gamma^n)$, hence all $c(E)$, uniquely. The same argument proves uniqueness of the Stiefel–Whitney and Pontryagin classes. ■

Remark 5 (Why Chern–Weil cannot see torsion). The Chern–Weil homomorphism lands in *real* de Rham cohomology, which is torsion-free and hence cannot detect classes like w_1, w_2 that live in mod-2 cohomology. The classifying-space point of view repairs this: it defines the integral classes $c_i \in H^*(\mathrm{BU}; \mathbb{Z})$ and the mod-2 classes $w_i \in H^*(\mathrm{BO}; \mathbb{Z}/2)$ directly, and the Chern–Weil construction is recovered as the image of c_i in $H^*(\mathrm{BU}; \mathbb{R})$. This is the precise division of labour between Chapters 3–6 (geometric representatives) and this chapter (integral/torsion classes).

8.3 Schubert cells

The cohomology of Theorem 3 is computed from a CW structure on the Grassmannian. We describe it for $\mathrm{Gr}_k(\mathbb{C}^N)$, following the standard Schubert calculus.

Fix a complete flag

$$0 = F_0 \subset F_1 \subset \dots \subset F_N = \mathbb{C}^N, \quad \dim F_i = i.$$

For a k -subspace $W \subseteq \mathbb{C}^N$ and $1 \leq i \leq k$ define the *Schubert rank* $r_i(W) = \min\{j : \dim(W \cap F_j) = i\}$, and form the sequence $1 \leq r_1 < \dots < r_k \leq N$. Equivalently, associate the partition $\lambda = (\lambda_1 \geq \dots \geq \lambda_k \geq 0)$ with $\lambda_{k-i+1} = N - k + i - r_i$.

Definition 6 (Schubert cell). The *Schubert cell* $e_\lambda \subseteq \mathrm{Gr}_k(\mathbb{C}^N)$ indexed by the partition λ (contained in the $k \times (N - k)$ rectangle) is

$$e_\lambda = \{W \in \mathrm{Gr}_k(\mathbb{C}^N) : r_i(W) = N - k + i - \lambda_{k-i+1} \text{ for all } i\}.$$

Each e_λ is homeomorphic to an open cell $\mathbb{C}^{d(\lambda)}$ of complex dimension $d(\lambda) = |\lambda| = \sum_i \lambda_i$, and the e_λ form a CW decomposition of $\mathrm{Gr}_k(\mathbb{C}^N)$ with cells only in even real dimensions. The closure $\bar{e}_\lambda$ is the *Schubert variety* $\{W : r_i(W) \geq N - k + i - \lambda_{k-i+1} \forall i\}$.

Proposition 7 (Cellular cohomology). The classes $[\bar{e}_\lambda] \in H^{2|\lambda|}(\mathrm{Gr}_k(\mathbb{C}^N); \mathbb{Z})$, for λ ranging over partitions in the $k \times (N - k)$ rectangle, form a $\mathbb{Z}$ -basis of the cohomology; all differentials in the cellular complex vanish because there are cells only in even dimensions.

Indication. The CW structure gives a cellular chain complex with one generator per cell in even degree; since there are no cells in odd degrees, all boundary maps vanish, so the cellular cohomology is free on the Schubert classes. Symbolically, the cellular chain complex is

$$\dots \longrightarrow C_{2k+2} \xrightarrow{0} C_{2k} \xrightarrow{0} C_{2k-2} \longrightarrow \dots, \quad C_{2k} = \mathbb{Z}\langle e_\lambda : |\lambda| = k \rangle,$$

so that $H^{2k} = \mathbb{Z}\langle [\bar{e}_\lambda] : |\lambda| = k \rangle$, all differentials being zero. ■

The classes $c_i = c_i(\gamma^k)$ of Theorem 3 are the Schubert classes of the *special* partitions $(1, 1, \dots, 1)$ (the i th Chern class is $[\bar{e}_{(1^i)}]$, the class of the Schubert variety of codimension $2i$); the ring structure is the classical Littlewood–Richardson rule. Passing to the limit $N \rightarrow \infty$ gives $H^*(\mathrm{BU}(k); \mathbb{Z}) = \mathbb{Z}[c_1, \dots, c_k]$.

8.4 Homotopy properties of classifying spaces

Proposition 8 (Properties of BG). The classifying space BG is uniquely determined up to weak homotopy equivalence by the following properties:

- (i) there is a universal principal G -bundle $EG \rightarrow BG$ with EG contractible;
- (ii) $\Omega BG \simeq G$ (the loop space of BG is homotopy equivalent to G);
- (iii) $\pi_n(BG) \cong \pi_{n-1}(G)$ for $n \geq 2$.

Proof. (i) is the definition ($EG = \text{St}_k(\mathbb{C}^\infty)$, the Stiefel manifold, is contractible). (ii) and (iii) follow from the long exact homotopy sequence of the fibration $G \rightarrow EG \rightarrow BG$: since $\pi_n(EG) = 0$ for all n , the connecting maps give $\pi_n(BG) \cong \pi_{n-1}(G)$. ■

Example 9 (Elementary classifying spaces). $\text{BU}(1) = \mathbb{C}P^\infty$ (since $\text{St}_1(\mathbb{C}^\infty) = S^\infty$ is contractible and the Hopf action of $\text{U}(1)$ gives $S^\infty/\text{U}(1) = \mathbb{C}P^\infty$); likewise $\text{BO}(1) = \mathbb{R}P^\infty$ and $\text{B}(\mathbb{Z}/2) = \mathbb{R}P^\infty$. More generally $H^*(\mathbb{C}P^\infty; \mathbb{Z}) = \mathbb{Z}[c_1]$ and $H^*(\mathbb{R}P^\infty; \mathbb{Z}/2) = \mathbb{Z}/2[w_1]$, in agreement with Theorem 3 for $n = 1$.

Remark 10 (The slogan). Characteristic classes are “cohomology classes of the classifying space”: a class of G -bundles is natural in the bundle iff it is pulled back from a class on BG . This is the content of Theorem 2 combined with the naturality of the Chern–Weil homomorphism, and it is the cleanest way to remember why there are exactly four families of characteristic classes: they are the cohomology of BO , BSO , BU and BSpin .

Example 11 (The classifying spaces of the classical groups). The low-degree cohomology of the classical classifying spaces is

$$H^*(\text{BO}(n); \mathbb{Z}/2) = \mathbb{Z}/2[w_1, \dots, w_n], \quad H^*(\text{BSO}(n); \mathbb{R}) = \mathbb{R}[p_1, \dots, p_{\lfloor n/2 \rfloor}] \text{ (plus } e \text{ if } n \text{ even),}$$

$$H^*(\text{BU}(n); \mathbb{Z}) = \mathbb{Z}[c_1, \dots, c_n], \quad H^*(\text{BSp}(n); \mathbb{Z}) = \mathbb{Z}[q_1, \dots, q_n],$$

where q_i are the *symplectic Pontryagin classes*. The vanishing $w_1|_{\text{BSO}} = 0$ and $w_2|_{\text{BSpin}} = 0$ encodes orientability and spin in the classifying-space language: $\text{BSO}(n) \rightarrow \text{BO}(n)$ is the double cover killing w_1 , and $\text{BSpin}(n) \rightarrow \text{BSO}(n)$ the double cover killing w_2 .

Remark 12 (The Leray–Serre spectral sequence). A second, more flexible route to the cohomology of BG is the Serre spectral sequence of the fibration $G \rightarrow EG \rightarrow BG$. Since EG is contractible, the spectral sequence has $E_2^{p,q} = H^p(BG; H^q(G))$ converging to $H^{p+q}(\text{pt})$, which forces $H^*(BG)$ to be dual to $H^*(G)$. For $G = \text{U}(n)$ this recovers $H^*(\text{BU}) = \mathbb{Z}[c_1, \dots, c_n]$ from $H^*(\text{U}(n)) = \Lambda(x_1, x_3, \dots, x_{2n-1})$ (the exterior algebra on odd generators), via the transgression which identifies c_i with the transgressions of the odd generators. This is the bridge between the de Rham generators of G and the characteristic classes.

9 Applications and Perspectives

9.1 The index theorems

The deepest applications of characteristic classes are to the index theorems, which assert that certain integers — analytically defined indices of elliptic operators — are computed by integrating characteristic classes. We state the prototypes.

Theorem 1 (Gauss–Bonnet–Chern). For a closed oriented manifold M of even dimension n ,

$$\chi(M) = \int_M e(TM) = \text{ind}(d + d^*),$$

where $d + d^*$ is the de Rham operator graded by form parity (even/odd).

Theorem 2 (Hirzebruch signature theorem). For a closed oriented $4k$ -manifold M ,

$$\text{sign}(M) = \int_M L_k(p_1, \dots, p_k),$$

where sign is the signature of the intersection form and L_k is the multiplicative L -genus, a polynomial in the Pontryagin classes.

Theorem 3 (Hirzebruch–Riemann–Roch). For a holomorphic vector bundle E over a compact complex manifold M ,

$$\chi(M, E) = \int_M \text{ch}(E) \text{td}(TM),$$

where $\chi(M, E) = \sum_i (-1)^i \dim H^i(M; \mathcal{O}(E))$ is the holomorphic Euler characteristic and td is the Todd class.

All three are special cases of the Atiyah–Singer index theorem

$$\text{ind}(D) = \int_M \text{ch}(\sigma(D)) \text{td}(TM),$$

and they exhibit the same template as Gauss–Bonnet: a characteristic class (the Euler class, the L -genus, $\text{ch} \cdot \text{td}$) integrated over M equals an analytic invariant. The Hirzebruch–Riemann–Roch theorem for a curve is precisely the classical Riemann–Roch theorem; for a surface it specialises to the Noether formula.

Theorem 4 (Atiyah–Singer index theorem). Let D be an elliptic differential operator on a closed manifold M , with symbol class $\sigma(D) \in K(TM)$. Then

$$\text{ind}(D) = \int_M \text{ch}(\sigma(D)) \text{td}(TM_{\mathbb{C}}).$$

The three theorems above are obtained by specialising D :

- for $D = d + d^*$ graded by form parity, $\text{ch}(\sigma(D)) \text{td} = e(TM)$, giving Gauss–Bonnet–Chern;
- for $D = d + d^*$ graded by the sign of the Hodge $*$ operator, the integrand is the L -genus, giving the signature theorem;
- for $D = \bar{\partial} + \bar{\partial}^*$ on a Kähler manifold, $\text{ch}(\sigma(D)) \text{td} = \text{ch}(E) \text{td}(TM)$, giving Hirzebruch–Riemann–Roch.

Remark 5 (The role of characteristic classes). What makes the Chern–Weil theory of Chapters 3–6 the indispensable tool of the index theorems is this: the right-hand side $\text{ch} \cdot \text{td}$ is, by construction, a *curvature differential form*, so it can be computed locally from the geometry. The characteristic classes are thus the bridge over which a global analytic

invariant (the index) is expressed as an integral of local geometric data. This is the deepest and most far-reaching application of the subject.

Remark 6 (*Unity of the four families*). The four families form a single ladder, each a refinement of the previous: Stiefel–Whitney classes are the mod-2 obstructions (orientability, spin); Chern classes are the integral classes of complex bundles; Pontryagin classes are the real classes obtained by complexification; and the Euler class is the orientation class, whose top-degree integral is the Euler characteristic. The relations $w_{2k} \equiv c_k \pmod{2}$, $p_k = (-1)^k c_{2k}(E_{\mathbb{C}})$, $e^2 = p_n$ and $e = c_1$ for complex line bundles exhibit how each family is determined by the ones below it.

9.2 Worked examples of the index theorems

Example 7 (Riemann–Roch for curves). For a holomorphic line bundle L of degree d over a compact Riemann surface Σ of genus g ,

$$\text{ch}(L) = 1 + c_1(L), \quad \text{td}(T\Sigma) = 1 + \frac{1}{2}c_1(T\Sigma) = 1 + \frac{1}{2}(2 - 2g)[\omega],$$

so the Hirzebruch–Riemann–Roch theorem gives

$$\chi(\Sigma, L) = \int_{\Sigma} (1 + c_1(L))(1 + \frac{1}{2}c_1(T\Sigma)) = d + 1 - g,$$

which is the classical Riemann–Roch formula $\dim H^0(\Sigma, L) - \dim H^1(\Sigma, L) = d + 1 - g$ (for $d > 2g - 2$ the H^1 term vanishes). This single computation links the Chern class, the Todd class and the Euler characteristic in the exact setting of the Riemann–Roch theorem.

Example 8 (The Noether formula for surfaces). For a compact complex surface S , the Hirzebruch–Riemann–Roch theorem for the structure sheaf gives

$$\chi(S, \mathcal{O}_S) = \int_S \text{td}(TS) = \frac{c_1^2 + c_2}{12} = \frac{p_1 + 3\chi(S)}{12}$$

(the Noether formula), where $c_1^2 = \int_S c_1(TS)^2$ and $c_2 = \chi(S)$ by Gauss–Bonnet. The two Chern numbers c_1^2 and c_2 thus determine the holomorphic Euler characteristic.

Example 9 (The L -genus and the signature of $K3$). A $K3$ surface has $c_1 = 0$, $c_2 = 24$, so $p_1 = -2c_2 = -48$ and $\int p_1 = -48$. The signature is $\text{sign}(K3) = \int p_1/3 = -16$, the famous value -16 of the $K3$ surface; this is a model instance of the Hirzebruch signature theorem.

Remark 10 (*The analytic side*). The index theorems assert equalities between a topological quantity (a characteristic number) and an analytic quantity (an index). The analytic side — heat-kernel asymptotics, the Atiyah–Singer local index formula — is beyond these notes; what the Chern–Weil theory provides is the *topological* side, i.e. the precise characteristic classes that enter. This is why the curvature representatives of Chapters 3–6 are exactly the ingredients of the index theorems: the integrand $\text{ch}(\sigma(D)) \text{td}(TM)$ is a curvature differential form by construction.

9.3 A historical sketch

The theory grew in four decisive steps. *Stiefel–Whitney* (1935–1940): the obstruction to linearly independent vector fields, reformulated by Whitney as cohomology classes. *Pontryagin* (1942): integral classes of real bundles via “characteristic cycles”. *Chern* (1944–1946): the intrinsic proof of the Gauss–Bonnet–Chern theorem (1944) and the definitive construction of the Chern classes through the curvature of a connection (1946). *Milnor–Stasheff* (1974): the axiomatic and classifying-space systematisation that made the subject a self-contained chapter of algebraic topology. The sequel — the index theorems of Hirzebruch (1954–56) and Atiyah–Singer (1963) — turned the characteristic classes from bookkeeping devices into the bridge between topology, geometry and analysis.

9.4 What we did not cover

This account has been confined to the de Rham and classifying-space points of view over $\mathbb{R}$, $\mathbb{Z}$ and $\mathbb{Z}/2$. A complete treatment would add: the obstruction-theoretic construction of the Stiefel–Whitney classes via Steenrod squares and the Wu classes; the full theory of cobordism (the Thom spectra, the Hirzebruch signature theorem in its integral form); the Chern–Simons *secondary* classes and their role in 3-manifold invariants; and the equivariant and infinite-dimensional versions that enter the fixed-point and localisation formulas. These all rest on the same Chern–Weil and classifying-space foundations developed above, and the reader is referred to the references for each.

9.5 Further reading

For a complete treatment of the geometric (Chern–Weil) side, Tu’s *Differential Geometry* (Chapters 20–23) is the closest to our exposition and is the recommended first reference. Bott–Tu’s *Differential Forms in Algebraic Topology* gives the de Rham and obstruction-theoretic viewpoints together with the Thom and Euler classes and the spectral-sequence computations. Milnor–Stasheff is the standard axiomatic and classifying-space reference, with the complete theory of Stiefel–Whitney, Chern and Pontryagin classes and of cobordism. For the index theorems, Hirzebruch’s *Topological Methods in Algebraic Geometry* and the Atiyah–Singer papers are the classical sources; Kobayashi–Nomizu Volume II supplies the principal-bundle foundations. The Chinese monographs 周建伟 《微分几何讲义》 and 梅加强 《流形与几何初步》 are concise companions for the geometric prerequisites.

9.6 Summary of the four families

We close with a table collecting the four families of characteristic classes, the structure group of each, and the geometric/topological content.

Class	Group	Degree	Content
$w_k(E)$	$O(n), \mathbb{Z}/2$	k	orientability, spin, Steenrod squares
$c_k(E)$	$U(n), \mathbb{Z}$	$2k$	complex bundles, divisors
$p_k(E)$	$SO(n), \mathbb{R}$	$4k$	real bundles, cobordism, L -genus
$e(E)$	$SO(2n), \mathbb{R}$	$2n$	orientation, Gauss–Bonnet

The relations

$$w_{2k} \equiv c_k \pmod{2}, \quad p_k = (-1)^k c_{2k}(E_{\mathbb{C}}), \quad e^2 = p_n, \quad e(L_{\mathbb{R}}) = c_1(L)$$

summarise how the four families interlock, and the index theorems $\text{ind} = \int \text{ch} \cdot \text{td}$ (with the Euler class, L -genus and $\text{ch} \cdot \text{td}$ as the three specialisations) exhibit how they enter geometry and analysis.

Appendix A Symmetric Polynomials and the Newton Identities

The algebra of invariant polynomials $I^*(\text{GL}(n))$ used throughout is the algebra of symmetric polynomials in n variables, and we record here the facts used in Chapters 3 and 4. Let $\Lambda_n = \mathbb{Z}[x_1, \dots, x_n]^{S_n}$ be the ring of symmetric polynomials in n variables.

Definition 1 (Elementary, complete and power sums). The *elementary symmetric polynomials* σ_k , the *power sums* p_k , and the *complete homogeneous symmetric polynomials* h_k are

$$\sigma_k = \sum_{i_1 < \dots < i_k} x_{i_1} \cdots x_{i_k}, \quad p_k = \sum_i x_i^k, \quad h_k = \sum_{i_1 \leq \dots \leq i_k} x_{i_1} \cdots x_{i_k},$$

with generating functions

$$\sum_k \sigma_k t^k = \prod_i (1 + x_i t), \quad \sum_k h_k t^k = \prod_i (1 - x_i t)^{-1}, \quad \sum_k p_k t^{k-1} = \sum_i \frac{x_i}{1 - x_i t}.$$

Theorem 2 (Fundamental theorem of symmetric polynomials). $\Lambda_n = \mathbb{Z}[\sigma_1, \dots, \sigma_n]$; that is, every symmetric polynomial is uniquely a polynomial in the elementary symmetric functions, which are algebraically independent.

Indication. Order monomials $x_1^{a_1} \cdots x_n^{a_n}$ lexicographically. The leading monomial of a symmetric polynomial f is $x_1^{a_1} \cdots x_n^{a_n}$ with $a_1 \geq \dots \geq a_n$; subtract an appropriate multiple of $\sigma_1^{a_1-a_2} \sigma_2^{a_2-a_3} \cdots \sigma_n^{a_n}$, which has the same leading monomial, and induct. Independence follows by counting degrees. Symbolically, writing $\text{LM}(f)$ for the leading monomial,

$$\text{LM}(\sigma_1^{a_1-a_2} \sigma_2^{a_2-a_3} \cdots \sigma_n^{a_n}) = x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n},$$

so $f - c \sigma_1^{a_1-a_2} \cdots \sigma_n^{a_n}$ has strictly smaller leading monomial, and descending induction terminates. $\blacksquare$

Proposition 3 (Newton identities). For $k \geq 1$,

$$k \sigma_k = \sum_{j=1}^k (-1)^{j-1} p_j \sigma_{k-j}, \quad k h_k = \sum_{j=1}^k p_j h_{k-j}.$$

Proof. The elementary symmetric polynomials have generating function $\sigma(t) = \sum_{k \geq 0} \sigma_k t^k = \prod_i (1 + x_i t)$ with $\sigma_0 = 1$. Taking the logarithmic derivative,

$$\frac{\sigma'(t)}{\sigma(t)} = \sum_i \frac{x_i}{1 + x_i t} = \sum_i \sum_{j \geq 1} (-1)^{j-1} x_i^j t^{j-1} = \sum_{j \geq 1} (-1)^{j-1} p_j t^{j-1},$$

where the second equality is the geometric series $(1 + x_i t)^{-1} = \sum_{j \geq 1} (-1)^{j-1} (x_i t)^{j-1}$ and the third uses $p_j = \sum_i x_i^j$. Cross-multiplying,

$$\sum_{k \geq 1} k \sigma_k t^{k-1} = \left(\sum_{k \geq 0} \sigma_k t^k \right) \left(\sum_{j \geq 1} (-1)^{j-1} p_j t^{j-1} \right).$$

The coefficient of t^{k-1} on the right is the sum over the pairs (ℓ, j) with $\ell + (j - 1) = k - 1$, i.e. $j = k - \ell$ for $0 \leq \ell \leq k - 1$; hence

$$k \sigma_k = \sum_{\ell=0}^{k-1} (-1)^{k-\ell-1} p_{k-\ell} \sigma_\ell = \sum_{j=1}^k (-1)^{j-1} p_j \sigma_{k-j},$$

which is the first identity.

For the second, the complete symmetric polynomials have generating function $h(t) = \sum_{k \geq 0} h_k t^k = \prod_i (1 - x_i t)^{-1}$. The logarithmic derivative gives

$$\frac{h'(t)}{h(t)} = \sum_i \frac{x_i}{1 - x_i t} = \sum_i \sum_{j \geq 1} x_i^j t^{j-1} = \sum_{j \geq 1} p_j t^{j-1},$$

using $(1 - x_i t)^{-1} = \sum_{j \geq 1} (x_i t)^{j-1}$. Cross-multiplying and equating the coefficient of t^{k-1} yields

$$k h_k = \sum_{j=1}^k p_j h_{k-j},$$

the second identity. ■

Corollary 4. The ring $\Lambda_n \otimes \mathbb{Q} = \mathbb{Q}[\sigma_1, \dots, \sigma_n] = \mathbb{Q}[p_1, \dots, p_n]$, and the two sets $\{\sigma_k\}$ and $\{p_k\}$ are related by the Newton identities. This is the algebraic statement that $\{\text{tr}(A^k)\}$ and the coefficients of $\det(I + \frac{t}{2\pi i} A)$ generate the same algebra $I^*(\text{GL}(n))$ (Example 2).

Remark 5 (The splitting principle algebraically). The splitting principle (Theorem 9) says that any symmetric polynomial in the Chern roots x_i gives a characteristic class; the fundamental theorem says these are exactly the polynomials in the Chern classes. The two statements are the geometric and the algebraic faces of the same fact.

Appendix B Index of Notation

Symbol	Meaning
$M, \dim M = m$	smooth manifold, its dimension
$\Omega^k(M), \Omega^k(M; V)$	smooth k -forms, V -valued k -forms
$E \rightarrow M, \operatorname{rk} E = r$	vector bundle and its rank
$\Gamma(E), \mathfrak{X}(M)$	sections of E , vector fields on M
$\operatorname{End} E, \operatorname{Hom}(E, F)$	endomorphism and homomorphism bundles
$\nabla, \nabla_X s$	connection, covariant derivative
$\omega = (\omega_j^i), \Omega = (\Omega_j^i)$	connection and curvature forms
$R(X, Y)$	curvature tensor $\nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X, Y]}$
Γ_{ij}^k	Christoffel symbols
$P \rightarrow M, G$	principal G -bundle and structure group
$\mathfrak{g}, \operatorname{Ad}, \operatorname{ad}$	Lie algebra of G ; adjoint reps. of G and of $\mathfrak{g}$
$A^\#, H_p, V_p$	fundamental vertical field, horizontal/vertical subspaces
$I^k(G), I^*(G)$	invariant polynomials
$w : I^*(G) \rightarrow H^{2*}(M)$	Chern–Weil homomorphism
$c_k(E), c(E)$	Chern classes, total Chern class
$\operatorname{ch}(E)$	Chern character
$p_k(E), p(E)$	Pontryagin classes
$\operatorname{Pf}(A)$	Pfaffian
$e(E)$	Euler class
$w_k(E), w(E)$	Stiefel–Whitney classes
σ_k, p_k, h_k	elementary, power-sum, complete symmetric polynomials
$\operatorname{Gr}_k(\mathbb{F}^N), \gamma^k$	Grassmannian and tautological bundle
BG, EG	classifying space, universal bundle
$\chi(M)$	Euler characteristic
$[\omega], x = c_1(\mathcal{O}(1))$	de Rham class, hyperplane class

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